



Footprinting & Reconnaissance

Certified Ethical Hacking

Module 1

Author: [Tayyab Shahzad](#)

Supervisor: [Muhammad Azam](#)

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Introduction to Footprinting

Footprinting is the **first and one of the most critical stages in ethical hacking and penetration testing**. During this phase, a cybersecurity professional collects as much publicly available information as possible about a target organization before launching any type of attack or security assessment. The main purpose of footprinting is to understand the **digital presence, infrastructure, and external exposure of the target**.

In ethical hacking practices, footprinting techniques can be categorized into two types:

- **Passive Footprinting** – collecting information without interacting directly with the target systems.
- **Active Footprinting** – interacting with the target network to obtain information.

1.1 Overview of Footprinting

Footprinting refers to the systematic process of gathering detailed information about a target organization to understand its **online presence and external attack surface**.

During this process, security analysts collect various types of information including:

- *Domain names and subdomains*
- *Public IP addresses*
- *DNS records*
- *Network architecture*
- *Operating systems used*

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Footprinting Through Search Engines

Footprinting Through Search Engines

Search engines are powerful tools for collecting information about organizations from publicly available sources on the internet. Ethical hackers often use advanced search techniques to locate sensitive files, login portals, exposed directories, and other valuable information.

Popular search engines used for footprinting include:

- Google
- Bing
- Yahoo
- DuckDuckGo

Task 01: Footprinting using Advanced Google Hacking Techniques

Advanced Google Hacking involves using special search operators to extract hidden or sensitive information from search engine results.

These operators help cybersecurity professionals discover:

- Login portals
- Configuration files
- Public documents
- Sensitive directories
- Database backups

In this lab exercise, **EC-Council** was used as the target organization.

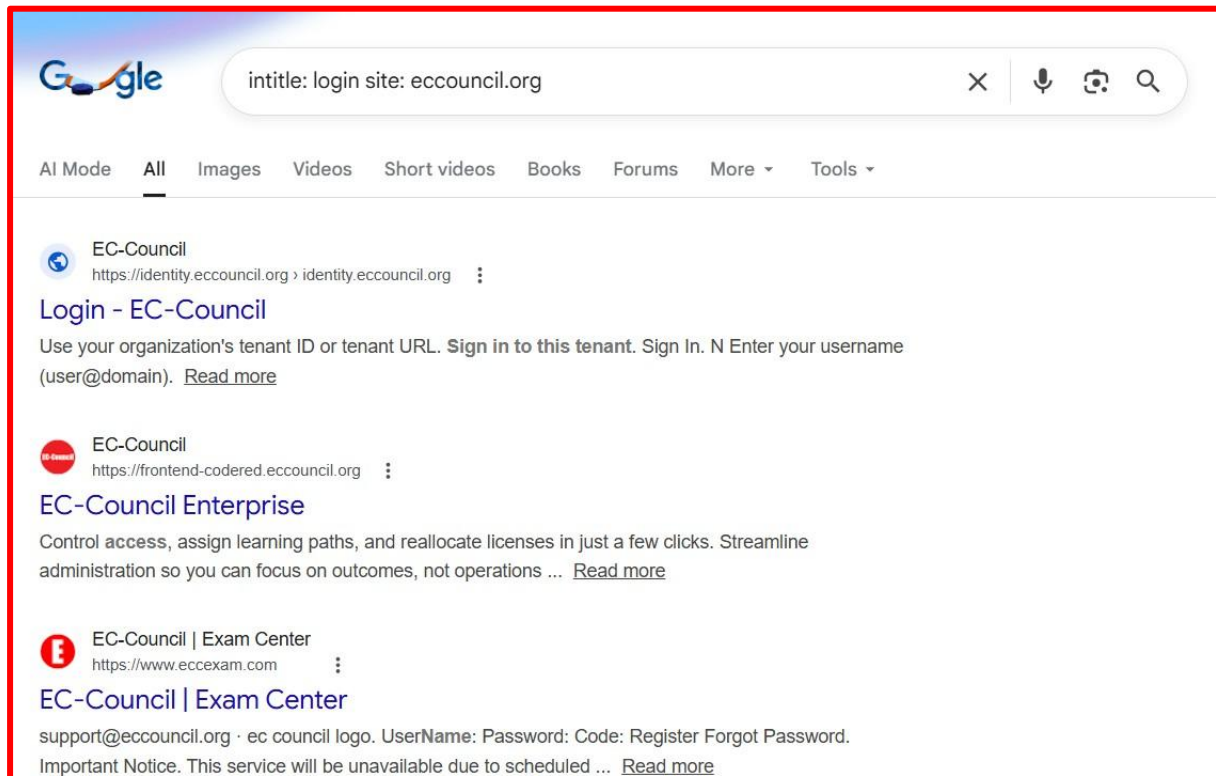
Step 1: Open Browser

Start the **Windows 10 virtual machine** and open the **Google Chrome browser**.

Step 2: Find Login Pages

In the Google search bar, enter the following command:

intitle:login site:eccouncil.org



Step 3: Search for Documents

Next, search for publicly available documents using:

[filetype:pdf CEH site:eccouncil.org](#)

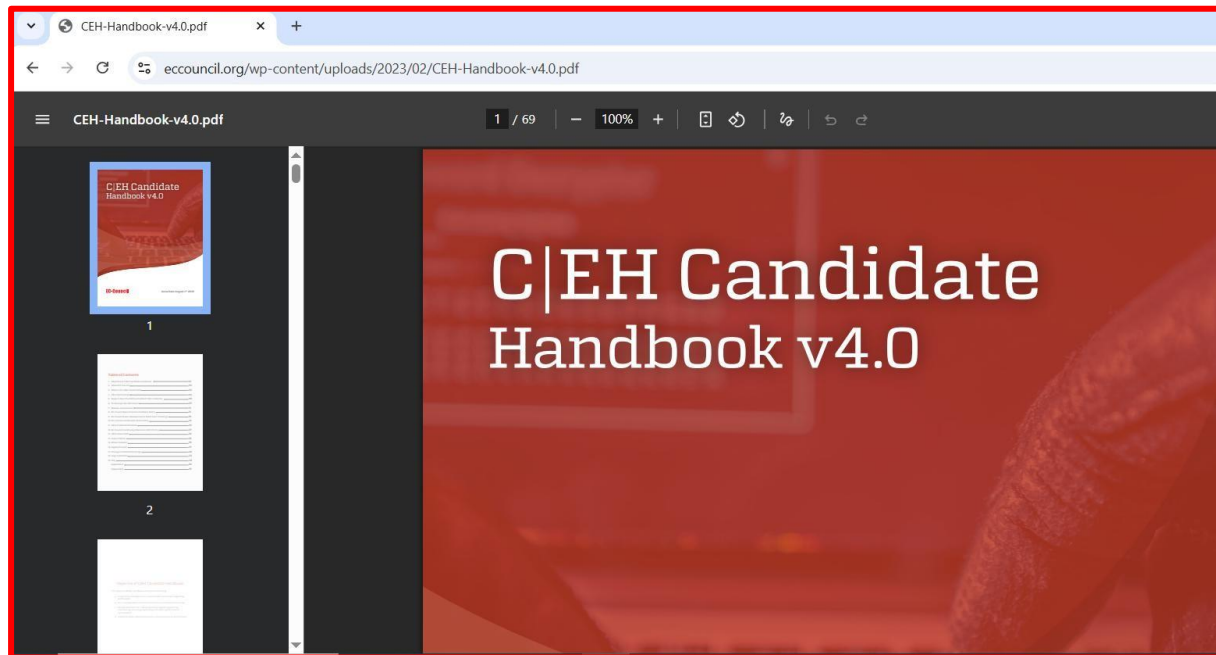
This query filters search results to display **PDF files related to CEH on the EC-Council website.**

Public documents may sometimes contain sensitive information such as:

- **Internal guidelines**
- **Technical documentation**
- **Organizational structure**
- **System configurations**

Step 4: Open a Document

Click any result such as **Candidate Handbook v4.0** to view the



Task 02: Information Gathering through Video Search Engines

Video search engines are specialized search platforms that allow users to locate and analyze video content available on the internet. From a cybersecurity perspective, video platforms can reveal useful information about organizations such as:

- Internal infrastructure glimpses
- Employee details
- Office locations
- Technologies being used
- Event recordings and training sessions

Such information can sometimes unintentionally expose sensitive organizational details.

In this lab, we used **YouTube** and a **YouTube Metadata analysis tool** to extract information related to a target organization.

Step 1: Open YouTube

Start the **Windows virtual machine** and open a web browser such as

Mozilla Firefox or Chrome.

In the browser address bar, navigate to:

<https://www.youtube.com>

The YouTube homepage will appear.

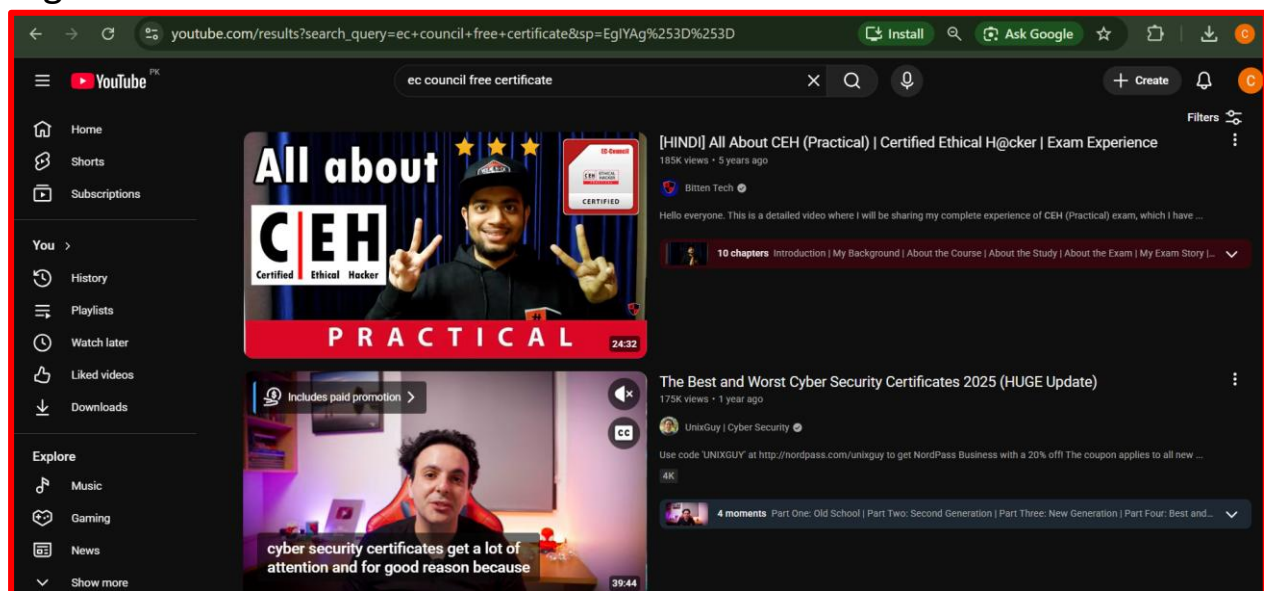
Step 2: Search for the Target Organization

In the YouTube search field, enter the name of the target organization.

In this lab example, the target organization was:

EC-Council

The search results will display videos uploaded by or related to the organization.



These videos may contain:

- *Conferences*
- *Training sessions*
- *Product demonstrations*
- *Organizational announcements*

All of these can provide valuable intelligence.

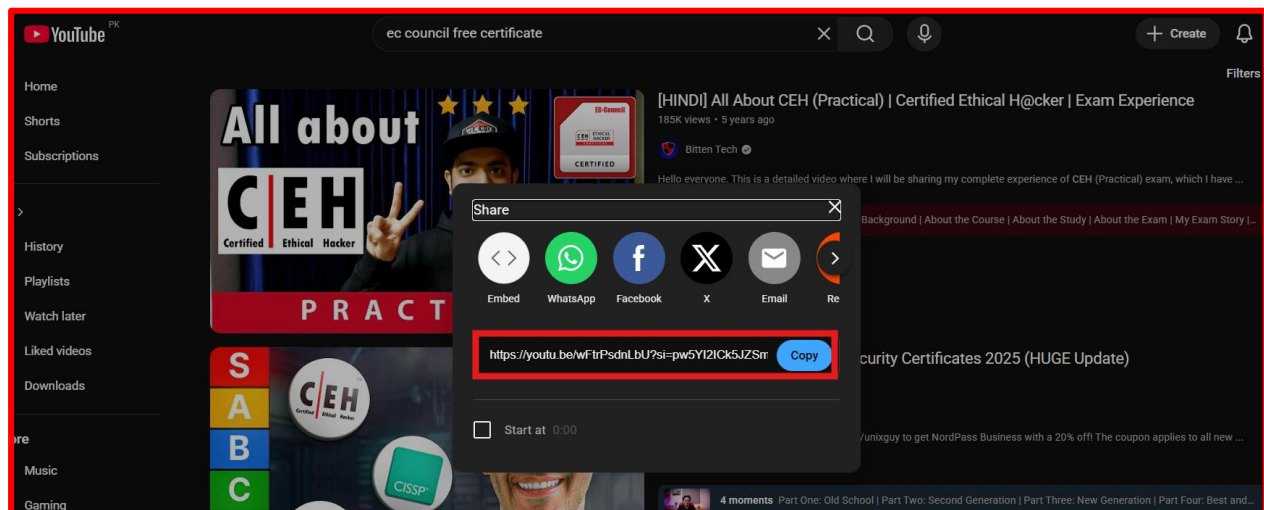
Step 3: Copy the Video Link

Select any video from the results.

Right-click on the video title and select

Copy Link

This copies the video URL to the clipboard.

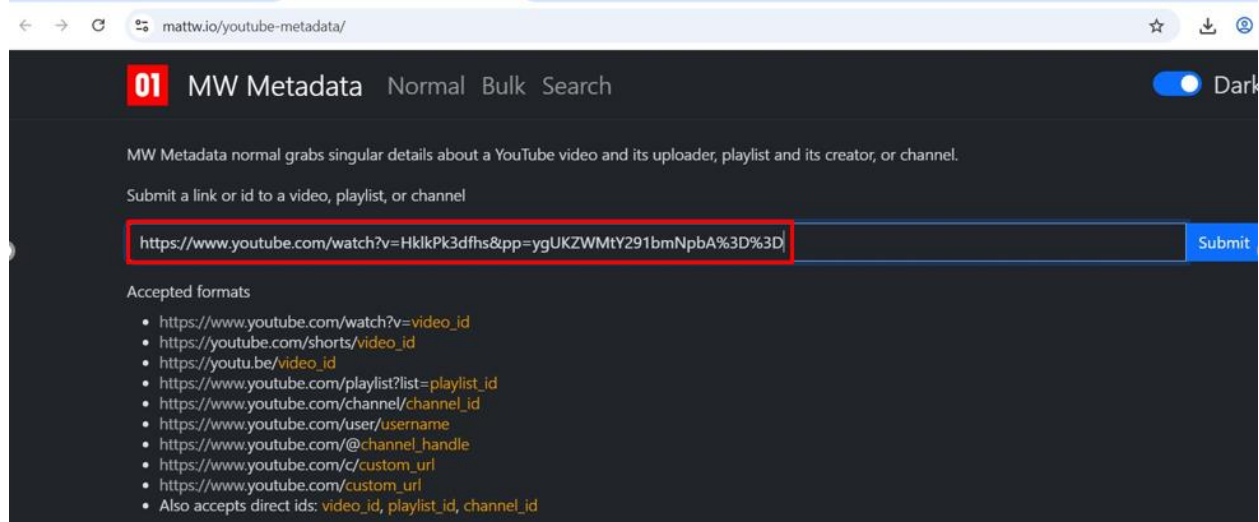


Step 4: Analyze Video Metadata

Open a new browser tab and visit the following metadata analysis tool:

<https://mattw.io/youtube-metadata/>

Paste the copied video link into the **Submit a link** field and click **Submit**.

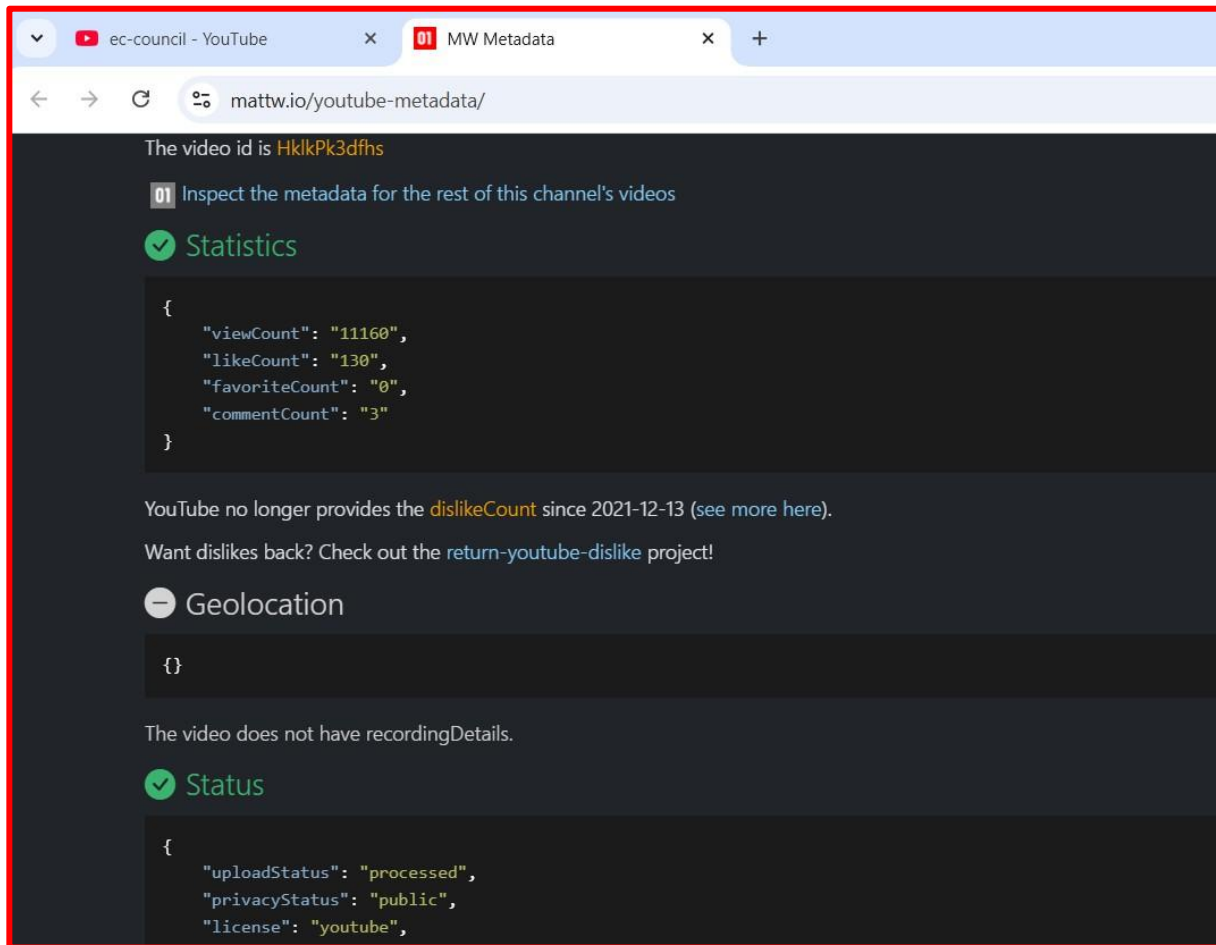


Step 5: Review Extracted Metadata

After submitting the link, the tool displays several important pieces of information including:

- **Video title**
- **Channel ID**
- **Upload date and time**
- **Video description**
- **Geolocation data (if available)**
- **View statistics**

This information is shown in different sections such as:



Step 6: Perform Reverse Image Search

Scroll to the **Thumbnail** section where thumbnails of the video are displayed.

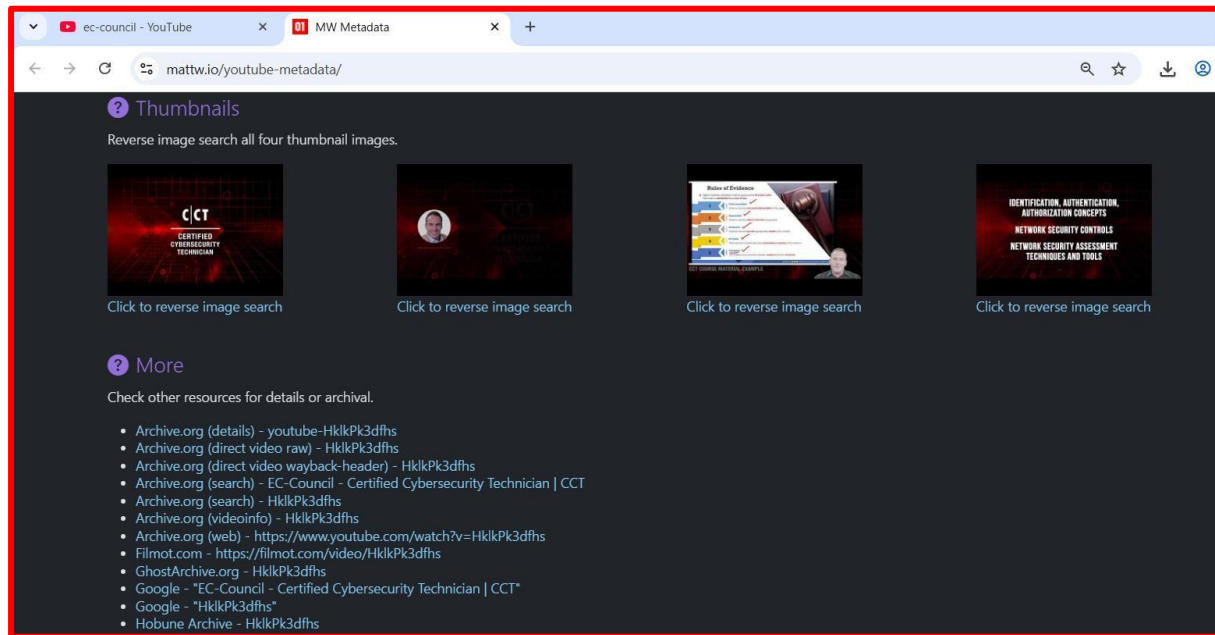
Click on **Reverse Image Search**.

This action opens Google Image Search to check if the same thumbnail appears elsewhere on the internet.

[Insert Screenshot: Reverse Image Search Result]

Reverse image searches can sometimes reveal:

- *Older versions of the video*
- *Related events*
- *Other websites hosting the same media*



Additional Tools for Video Intelligence

Other platforms that can be used for similar investigations include:

Video search engines:

- **Google Videos**
- **Yahoo Video Search**

Video analysis tools:

- **EZGif**
- **VideoReverser**

Reverse image tools:

- **TinEye**
- **Google Image Search**

These tools assist investigators in extracting hidden information from multimedia content.

Task 03: Gathering Information using FTP Search Engines

FTP (File Transfer Protocol) servers are commonly used by organizations to store and share large files such as software packages, backups, and internal documents.

Examples of information that may be discovered include:

- *Confidential documents*
- *Financial reports*
- *Internal software*
- *Employee records*
- *System backups*

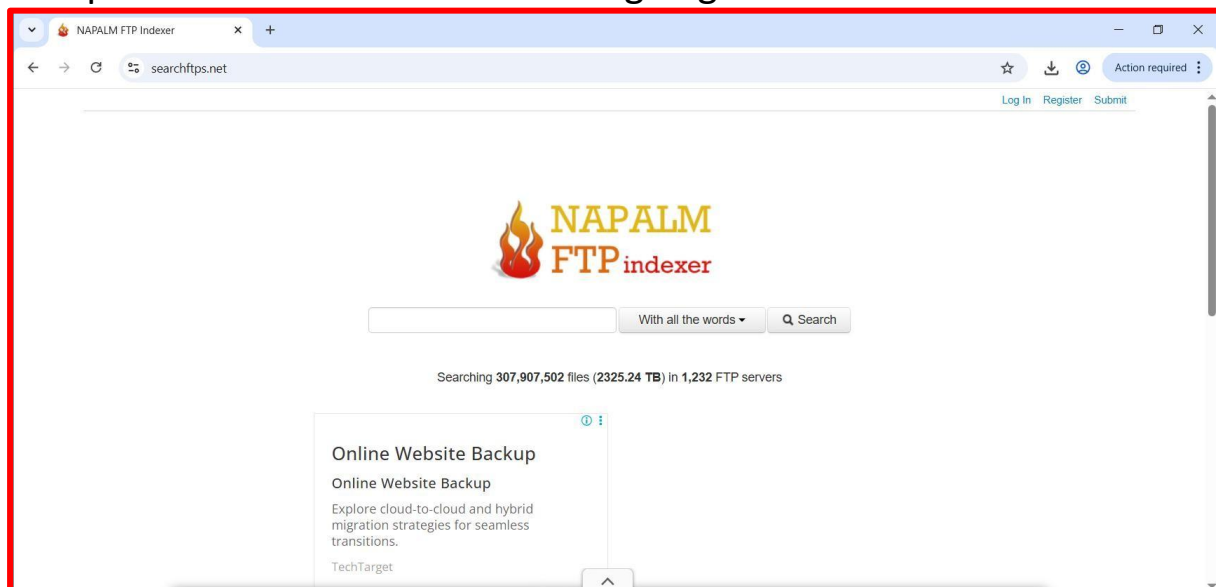
In this task, we used an FTP search engine to identify publicly accessible files related to a target organization.

Step 1: Open the FTP Search Engine

Launch a browser and visit the following website:

<https://www.searchftps.net/>

This platform acts as an FTP indexing engine.

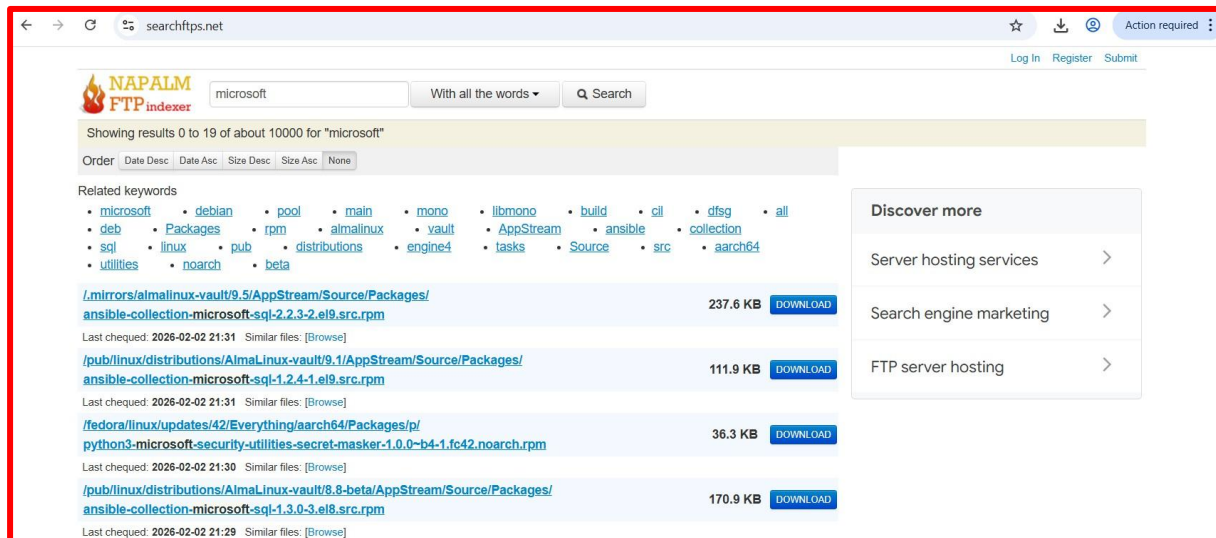


Step 3: Analyze the Results

The search engine displays files located on FTP servers associated with the target.

These files may include:

- *Software packages*
- *Documentation*
- *Backup files*
- *Configuration files*



Task 04: Information Gathering using IoT Search Engines

IoT (Internet of Things) search engines scan the internet to identify connected devices such as:

- *Smart cameras*
- *Routers*
- *Industrial controllers*
- *Traffic control systems*
- *Smart home devices*
- *SCADA systems*

These search engines index devices that expose services to the internet. Examples of IoT search engines include:

- **Shodan**
- **Censys**

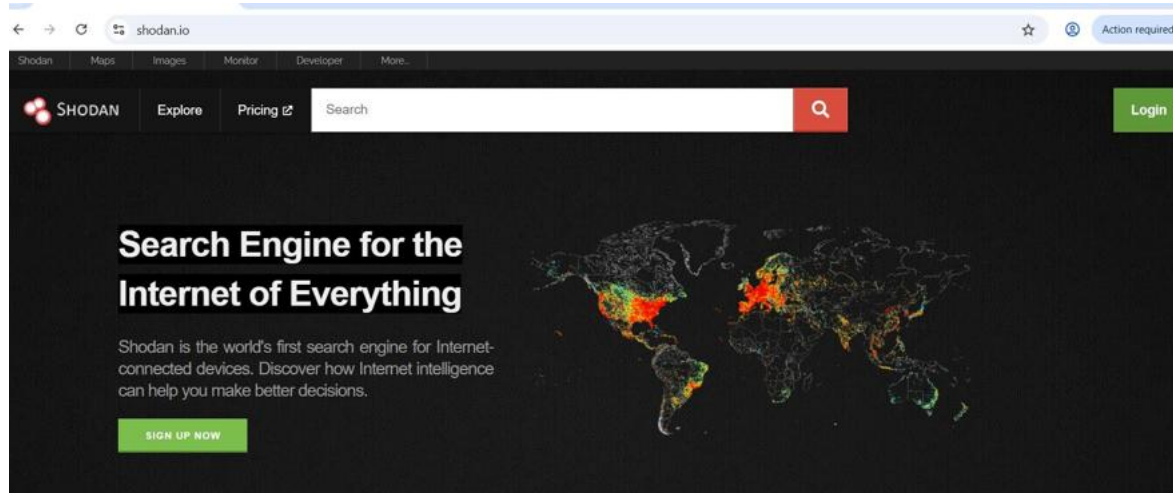
Such platforms are widely used in cybersecurity research to identify vulnerable devices.

Step 1: Open Shodan

Open a browser and navigate to:

<https://www.shodan.io>

Shodan's interface allows users to search for internet-connected devices worldwide.



Step 2: Search for Target Devices

In the search bar, enter the organization name.

Example:

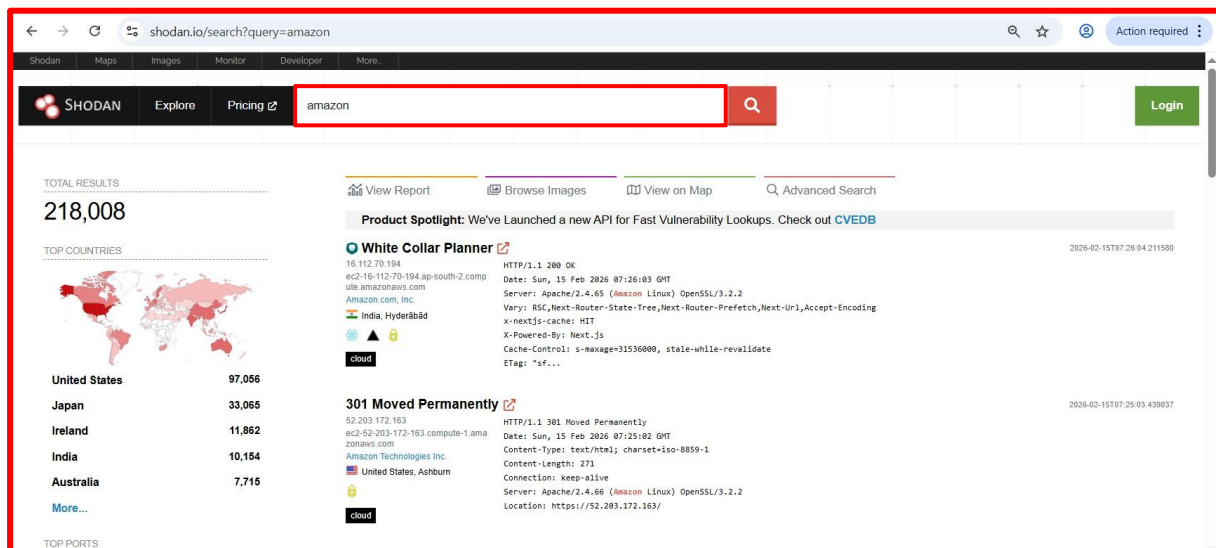
amazon

Press **Enter** to initiate the search.

Step 3: Review Device Information

Shodan will display a list of devices along with detailed information such as:

- *IP address*
- *Device type*
- *Operating system*
- *Open ports*
- *Location*
- *Service banners*



3

Footprinting through Internet Research Services

Footprinting through Internet Research Services

Internet Research Services are online platforms that provide publicly accessible information about individuals, companies, and network infrastructure. These services collect and organize data from various sources including web records, domain databases, social platforms, and archived internet pages.

From a cybersecurity perspective, these platforms can be extremely useful for **Open-Source Intelligence (OSINT)**. Ethical hackers often rely on them to collect important details about a target organization such as:

- *Domain information*
- *Subdomains and hosts*
- *Infrastructure technologies*
- *Physical locations*
- *Employee details*
- *Contact information*

Task 01: Identifying Domains, Subdomains, and Hosting Information Using Netcraft and DNSDumpster

Domains and subdomains are important components of an organization's digital infrastructure. By identifying these elements, ethical hackers can understand how a company structures its online services.

Tools like **Netcraft** and **DNSDumpster** allow investigators to discover:

- *Subdomains*
- *Hosting providers*
- *Network infrastructure*
- *DNS records*
- *Server technologies*

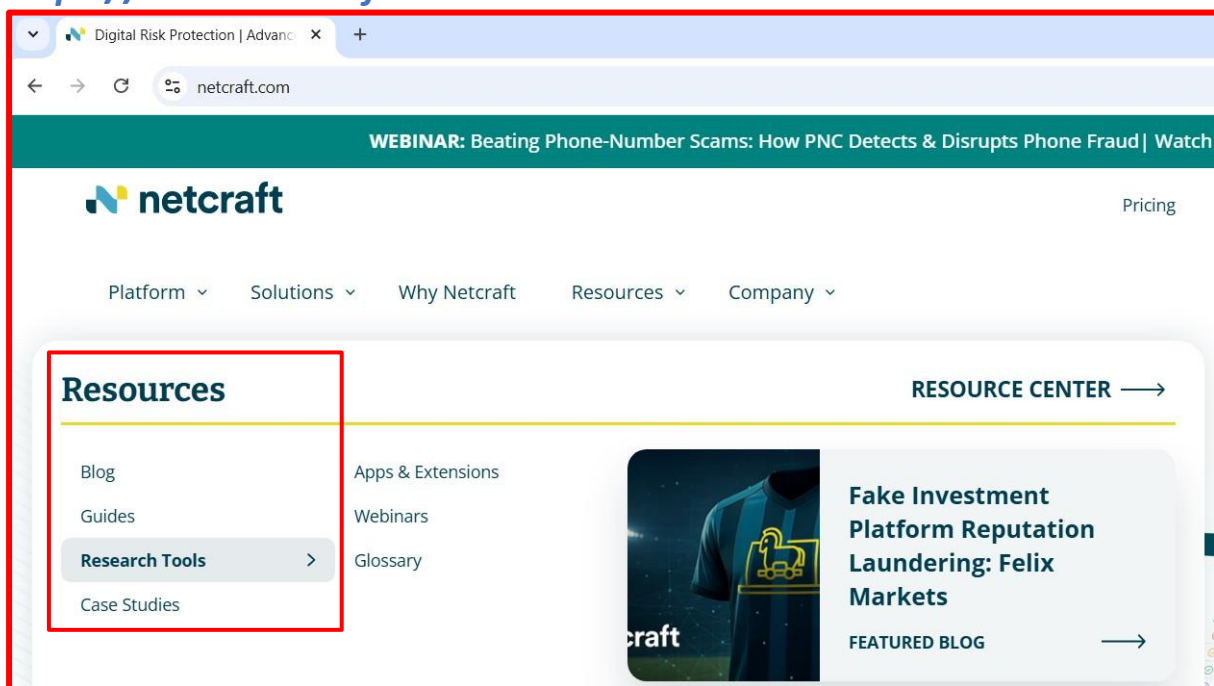
These tools help create a clearer picture of the target's online footprint.

Step 1: Access Netcraft

Start the **Windows virtual machine** and open any web browser.

Navigate to:

<https://www.netcraft.com>

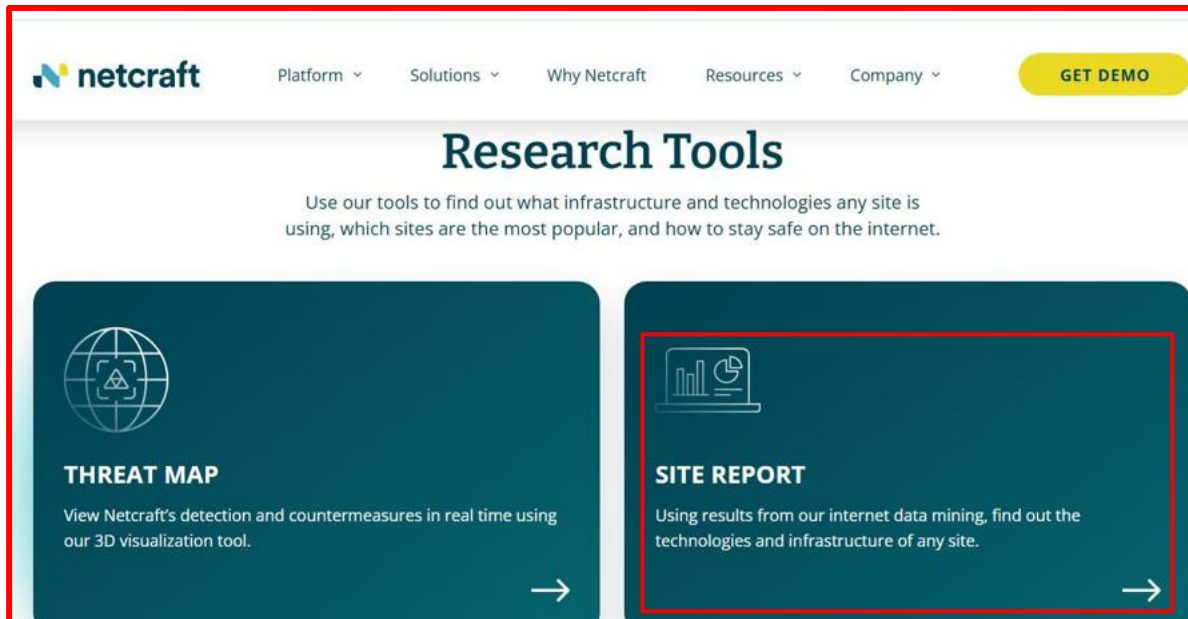


Step 2: Open Netcraft Research Tools

Click on the **menu icon** in the top-right corner.

Navigate to:

Resources → Research Tools
Then select **Site Report**.



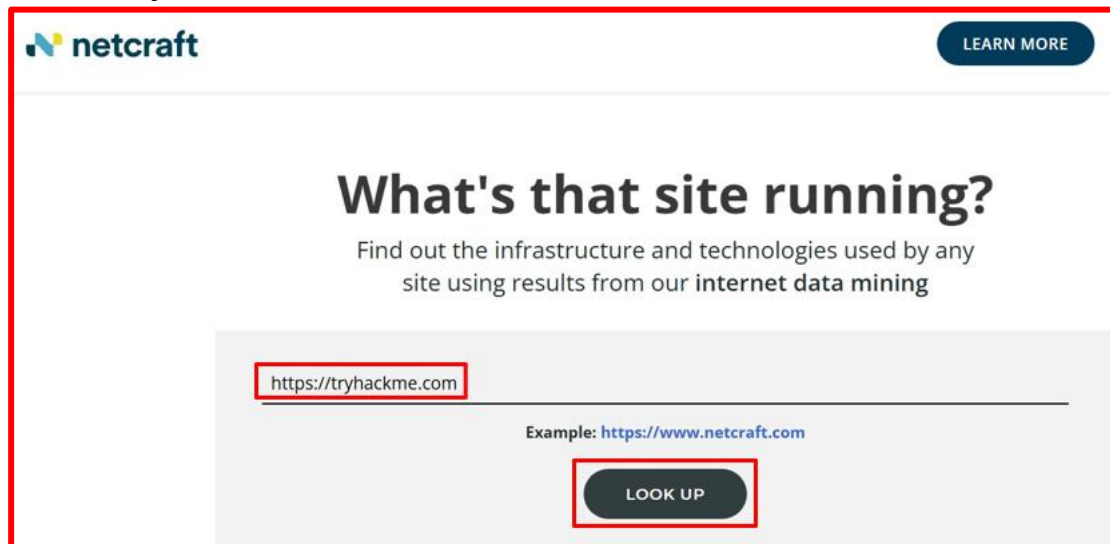
Step 3: Search for Target Website

Enter the target domain in the search field.

Example:

<https://tryhackme.com>

Click **Look Up**.



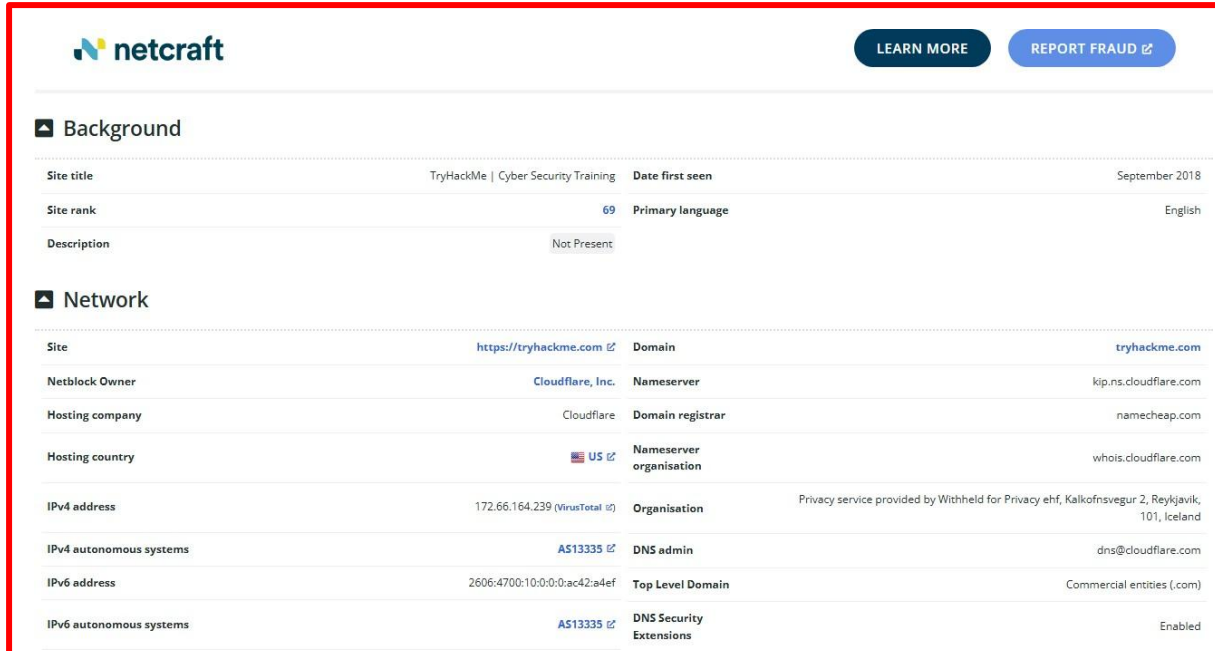
Step 4: Analyze Site Report

The Netcraft Site Report displays multiple sections including:

- *Background information*
- *Network details*

- **Hosting history**
- **Server technologies**

These details provide insights into how the target website is configured and hosted.



The screenshot shows the Netcraft website analysis tool interface. It features a 'Background' section with site details and a 'Network' section with technical information. The 'Background' section includes fields for Site title, Site rank, Description, Date first seen, Primary language, and a 'Not Present' status for Description. The 'Network' section lists Site, Netblock Owner, Hosting company, Hosting country, IPv4 address, IPv4 autonomous systems, IPv6 address, IPv6 autonomous systems, Domain, Nameserver, Domain registrar, Nameserver organisation, Organisation, DNS admin, Top Level Domain, and DNS Security Extensions.

Background	
Site title	TryHackMe Cyber Security Training
Date first seen	September 2018
Site rank	69
Primary language	English
Description	Not Present

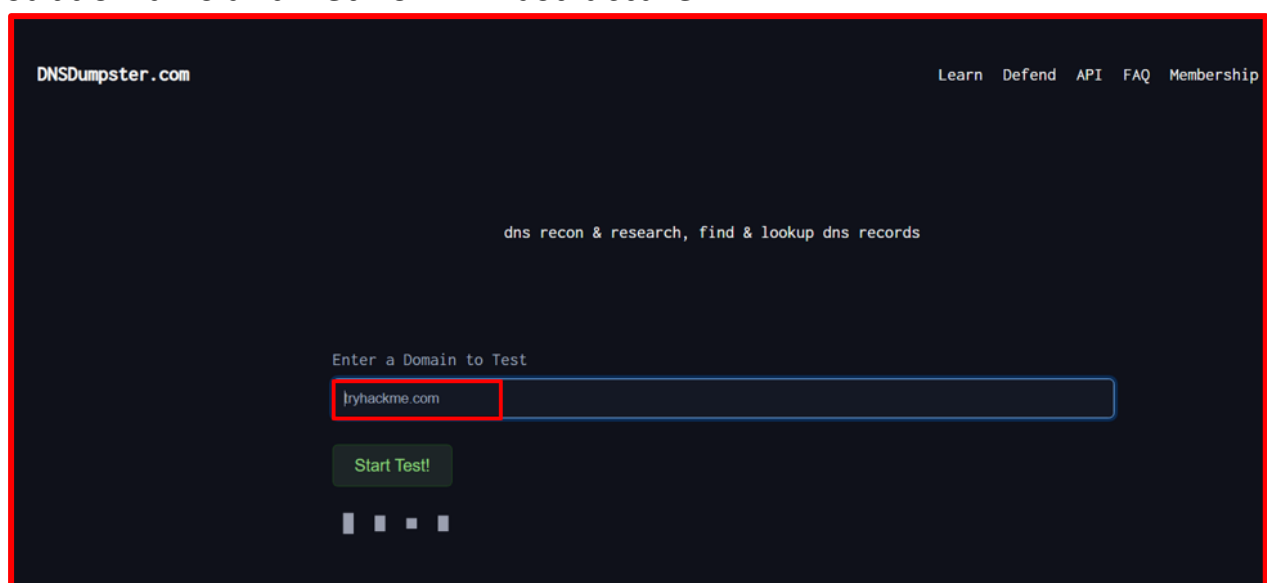
Network	
Site	https://tryhackme.com
Domain	tryhackme.com
Netblock Owner	Cloudflare, Inc.
Nameserver	kip.ns.cloudflare.com
Hosting company	Cloudflare
Domain registrar	namecheap.com
Hosting country	US
Nameserver organisation	whois.cloudflare.com
Organisation	Privacy service provided by Withheld for Privacy ehf, Kalkofnsvegur 2, Reykjavik, 101, Iceland
IPv4 address	172.66.164.239 (VirusTotal)
DNS admin	dns@cloudflare.com
IPv4 autonomous systems	AS13335
Top Level Domain	Commercial entities (.com)
IPv6 address	2606:4700:10:0:0:ac42:a4ef
DNS Security Extensions	Enabled
IPv6 autonomous systems	AS13335

Step 5: Use DNSDumpster for DNS Mapping

Open a new browser tab and visit:

<https://dnsdumpster.com>

DNSDumpster is a DNS reconnaissance tool that helps discover subdomains and network infrastructure.



The screenshot shows the DNSDumpster.com website interface. It features a dark theme with a navigation bar at the top containing links for Learn, Defend, API, FAQ, and Membership. The main content area has a heading 'dns recon & research, find & lookup dns records' and a form labeled 'Enter a Domain to Test'. The domain 'tryhackme.com' is entered into the form, and a 'Start Test!' button is visible below it.

Step 6: Search for Target Domain

Enter the target domain:

tryhackme.com

Click **Search**.



Step 7: Review DNS Information

DNSDumpster displays several important details including:

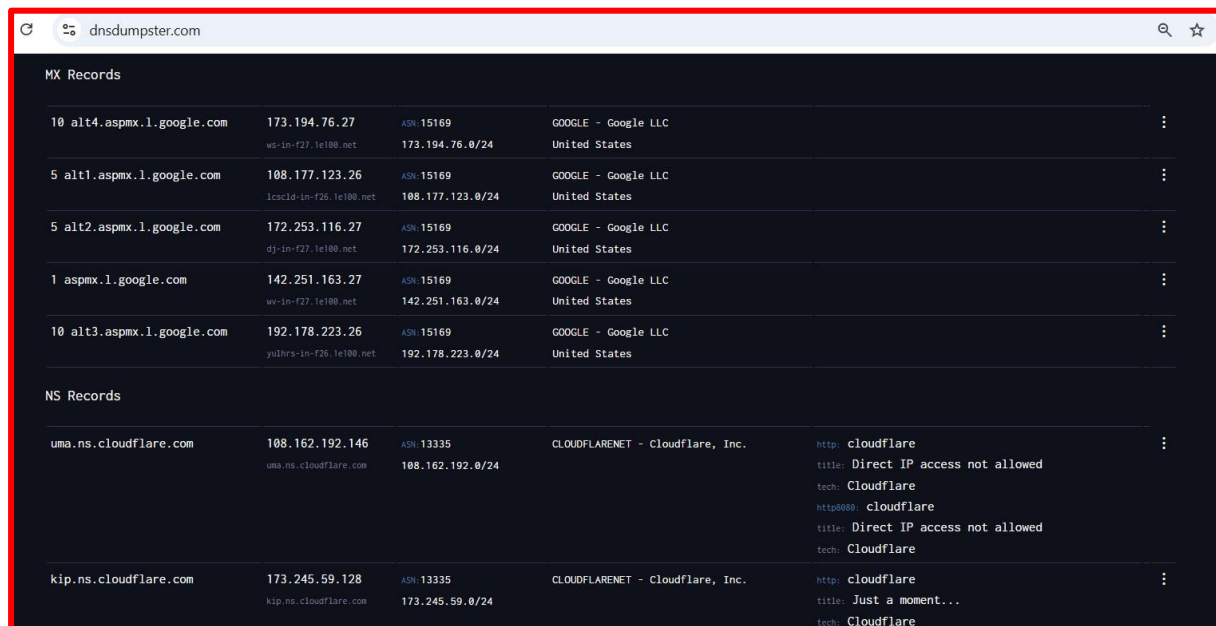
- **DNS Servers**
- **MX Records (Mail servers)**
- **Host Records (A records)**
- **IP addresses**

Step 8: View Domain Mapping

Scroll down to view the **domain mapping diagram** which visually represents the infrastructure of the domain.

The screenshot shows the 'A Records (subdomains from dataset)' section of dnsdumpster.com. It contains a table with columns: Host, IP, ASN, ASN Name, Open Services (from DB), and RevIP.

Host	IP	ASN	ASN Name	Open Services (from DB)	RevIP
admin.tryhackme.com	104.20.29.66	ASN: 13335 104.20.16.0/20	CLOUDFLARENET - Cloudflare, Inc.	http: cloudflare title: Direct IP access not allowed tech: Cloudflare https: cloudflare title: 403 Forbidden cn: ssl900826.cloudflaressl.com tech: Cloudflare http8080: cloudflare title: Direct IP access not allowed tech: Cloudflare	6
assets.tryhackme.com	65.8.54.71	ASN: 16509 65.8.48.0/21	AMAZON-02 - Amazon.com, Inc. United States	http: CloudFront title: ERROR: The request could not be sati tech: Amazon Web Services Amazon CloudFront	1325
auth.tryhackme.com	104.18.35.233	ASN: 13335 104.18.35.0/24	CLOUDFLARENET - Cloudflare, Inc.	http: cloudflare title: Direct IP access not allowed tech: Cloudflare http8080: cloudflare title: Direct IP access not allowed	880



The screenshot shows the dnsdumpster.com website. It displays MX records for google.com and NS records for cloudflare.com. The MX records list several mail exchangers for google.com, all pointing to IP addresses in the 173.194.76.0/24 range. The NS records list nameservers for cloudflare.com, all pointing to IP addresses in the 108.162.192.0/24 range.

Host	IP	ASN	Organization	Country
10 alt4.aspmx.l.google.com	173.194.76.27	AS15169	GOOGLE - Google LLC	United States
5 alt1.aspmx.l.google.com	108.177.123.26	AS15169	GOOGLE - Google LLC	United States
5 alt2.aspmx.l.google.com	172.253.116.27	AS15169	GOOGLE - Google LLC	United States
1 aspmx.l.google.com	142.251.163.27	AS15169	GOOGLE - Google LLC	United States
10 alt3.aspmx.l.google.com	192.178.223.26	AS15169	GOOGLE - Google LLC	United States

Host	IP	ASN	Organization	Country
uma.ns.cloudflare.com	108.162.192.146	AS13335	CLOUDFLARENET - Cloudflare, Inc.	United States
kip.ns.cloudflare.com	173.245.59.128	AS13335	CLOUDFLARENET - Cloudflare, Inc.	United States

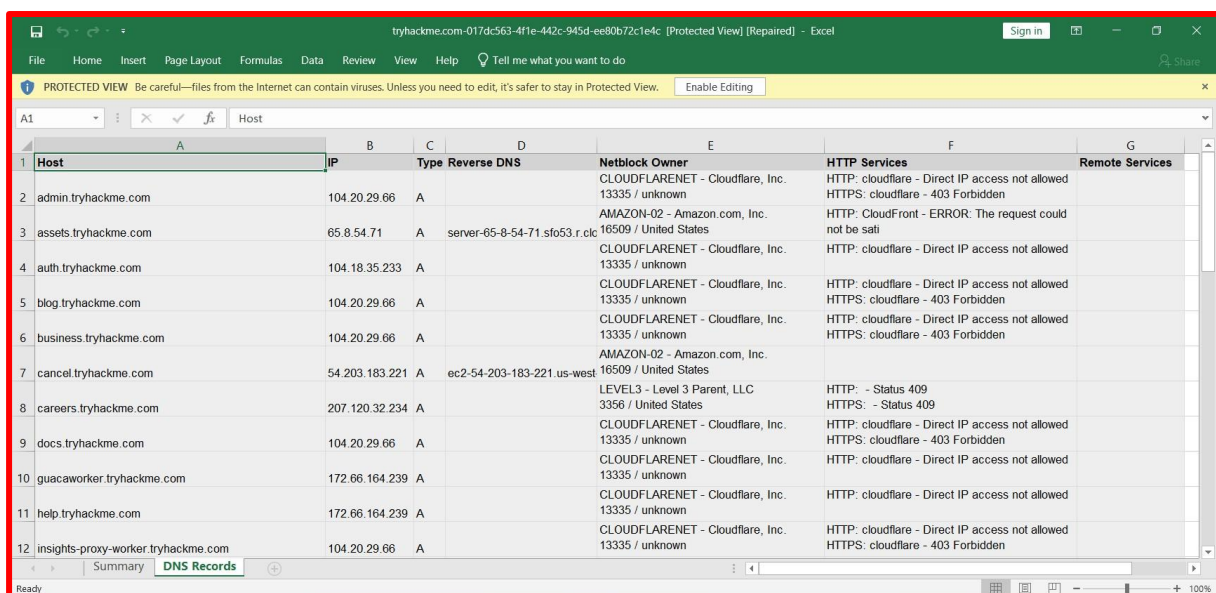
Step 9: Download Host Data

DNSDumpster also provides an option to download host information as an Excel file.

Click **Download .xlsx of Hosts**.

The downloaded file contains details such as:

- **Hostname**
- **IP address**
- **Country**
- **Reverse DNS**
- **Netblock owner**



The screenshot shows an Excel spreadsheet titled 'Host' with columns for Host, IP, Type, Reverse DNS, Netblock Owner, HTTP Services, and Remote Services. The data is organized into rows, with the first row being a header and the subsequent rows containing host information.

Host	IP	Type	Reverse DNS	Netblock Owner	HTTP Services	Remote Services
admin.tryhackme.com	104.20.29.66	A		CLOUDFLARENET - Cloudflare, Inc. 13335 / unknown	HTTP: cloudflare - Direct IP access not allowed HTTPS: cloudflare - 403 Forbidden	
assets.tryhackme.com	65.8.54.71	A	server-65-8-54-71.sfo53.r.cloud	AMAZON-02 - Amazon.com, Inc. 16509 / United States	HTTP: CloudFront - ERROR: The request could not be sat	
auth.tryhackme.com	104.18.35.233	A		CLOUDFLARENET - Cloudflare, Inc. 13335 / unknown	HTTP: cloudflare - Direct IP access not allowed	
blog.tryhackme.com	104.20.29.66	A		CLOUDFLARENET - Cloudflare, Inc. 13335 / unknown	HTTP: cloudflare - Direct IP access not allowed HTTPS: cloudflare - 403 Forbidden	
business.tryhackme.com	104.20.29.66	A		CLOUDFLARENET - Cloudflare, Inc. 13335 / unknown	HTTP: cloudflare - Direct IP access not allowed HTTPS: cloudflare - 403 Forbidden	
cancel.tryhackme.com	54.203.183.221	A	ec2-54-203-183-221.us-west	AMAZON-02 - Amazon.com, Inc. 16509 / United States		
careers.tryhackme.com	207.120.32.234	A		LEVEL3 - Level 3 Parent, LLC 3356 / United States	HTTP: - Status 409 HTTPS: - Status 409	
docs.tryhackme.com	104.20.29.66	A		CLOUDFLARENET - Cloudflare, Inc. 13335 / unknown	HTTP: cloudflare - Direct IP access not allowed HTTPS: cloudflare - 403 Forbidden	
guacaworker.tryhackme.com	172.66.164.239	A		CLOUDFLARENET - Cloudflare, Inc. 13335 / unknown	HTTP: cloudflare - Direct IP access not allowed	
help.tryhackme.com	172.66.164.239	A		CLOUDFLARENET - Cloudflare, Inc. 13335 / unknown	HTTP: cloudflare - Direct IP access not allowed	
insights-proxy-worker.tryhackme.com	104.20.29.66	A		CLOUDFLARENET - Cloudflare, Inc. 13335 / unknown	HTTP: cloudflare - Direct IP access not allowed HTTPS: cloudflare - 403 Forbidden	

Task 02: Gathering Personal Information using PeekYou

People search services are online platforms that collect public information about individuals from various sources including social media, public records, and websites.

These platforms can reveal information such as:

- Full names
- Addresses
- Email addresses
- Social media accounts
- Professional details
- Family connections

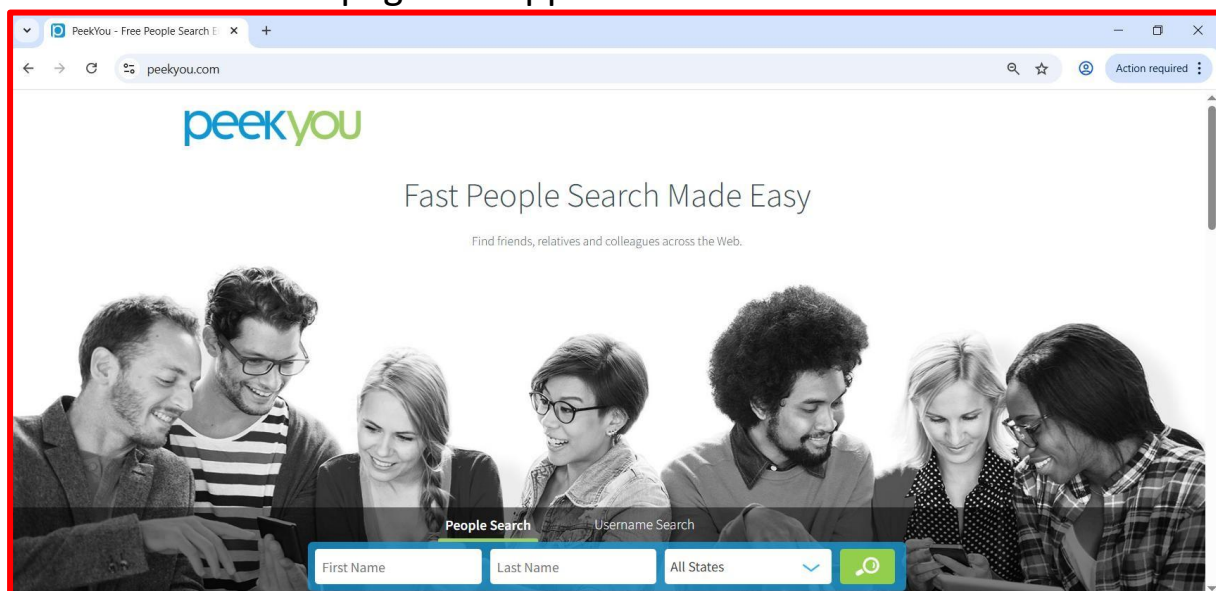
Attackers may use this information to conduct **social engineering** or **phishing attacks**.

Step 1: Open PeekYou

Open a browser and navigate to:

<https://www.peakyou.com>

The PeekYou homepage will appear.



Step 2: Search for a Person

Enter the following details in the search fields:

First Name:

Satya

Last Name:

Nadella

Location:

Washington

Click the **Search** button.

Step 3: Analyze the Results

The search results display various pieces of publicly available information including:

- **Public records**
- **Contact details**
- **Social media accounts**
- **Location history**
- **Associated profiles**

The screenshot shows the results of a search on peekyou.com for 'Satya Nadella' in Washington. The page is divided into several sections, each with a red box highlighting the section title. The 'Phonebook' section lists '1) Satya Nadella's Phone & Current Address' and '2) Social Media Profiles & More'. The 'Email Addresses' section lists several email addresses associated with Satya Nadella. The 'Contact Information & Locality History' section shows 'Satya N...' and a 'SEARCH DETAILS' button. The 'Facebook' section shows two profiles: 'Satya Nadella' and 'Satya Nadella (Sudama)'. The page also includes a 'Paid service sponsored by' notice for each section.

This information can potentially be misused by attackers to launch:

- **Social engineering attacks**
- **Identity theft**
- **Phishing campaigns**

Alternative People Search Tools

Other services that can be used for similar investigations include:

- **Spokeo**
- **Pipl**

- **Intelius**
- **BeenVerified**

These platforms aggregate data from multiple online sources to build detailed profiles.

Task 03: Email Harvesting using theHarvester

Emails are one of the most valuable pieces of information for attackers because they often serve as usernames for various services.

The **theHarvester** tool is widely used in penetration testing to collect:

- *Email addresses*
- *Subdomains*
- *Hostnames*
- *Employee names*
- *Open ports*

It gathers this information from multiple sources including search engines and public databases.

Step 1: Start Kali Linux

Turn on the **Kali Linux virtual machine**.

Open the **Terminal window**.

Step 2: Switch to Root User

Enter the following command:

sudo su

Password: **kali**

Step 3: Run theHarvester

Execute the following command:

theHarvester -d microsoft.com -l 200 -b duckduckgo

Where:

- **-d** specifies the target domain
- **-l** specifies the number of search results
- **-b** specifies the search engine source


```
root@kali: /home/kali
Session Actions Edit View Help

(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(kali@kali)-[/home/kali]
# theHarvester -d microsoft.com -l 200 -b duckduckgo

Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* theHarvester
*
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: microsoft.com
[*] Searching Duckduckgo.
[*] No IPs found.
[*] No emails found.
[*] No people found.
```

Step 4: Analyze Results

TheHarvester displays collected information such as:

- Email addresses
- Hostnames
- Subdomains

```
root@kali: /home/kali
Session Actions Edit View Help

[*] Hosts found: 27

2Fblogs.microsoft.com
2Fdocs.microsoft.com
2Fdownload.microsoft.com
2Fmsdn.microsoft.com
2Fnews.microsoft.com
Docs.microsoft.com
account.microsoft.com
appsource.microsoft.com
azure.microsoft.com
azuremarketplace.microsoft.com
blogs.microsoft.com
browserdefaults.microsoft.com
careers.microsoft.com
choice.microsoft.com
copilot.microsoft.com
developer.microsoft.com
docs.microsoft.com
download.microsoft.com
go.microsoft.com
```

These email addresses could later be used for:

- **Phishing attacks**
- **Credential brute-force attacks**
- **Social engineering campaigns**

Task 04: Determining Target Operating System using Passive Footprinting

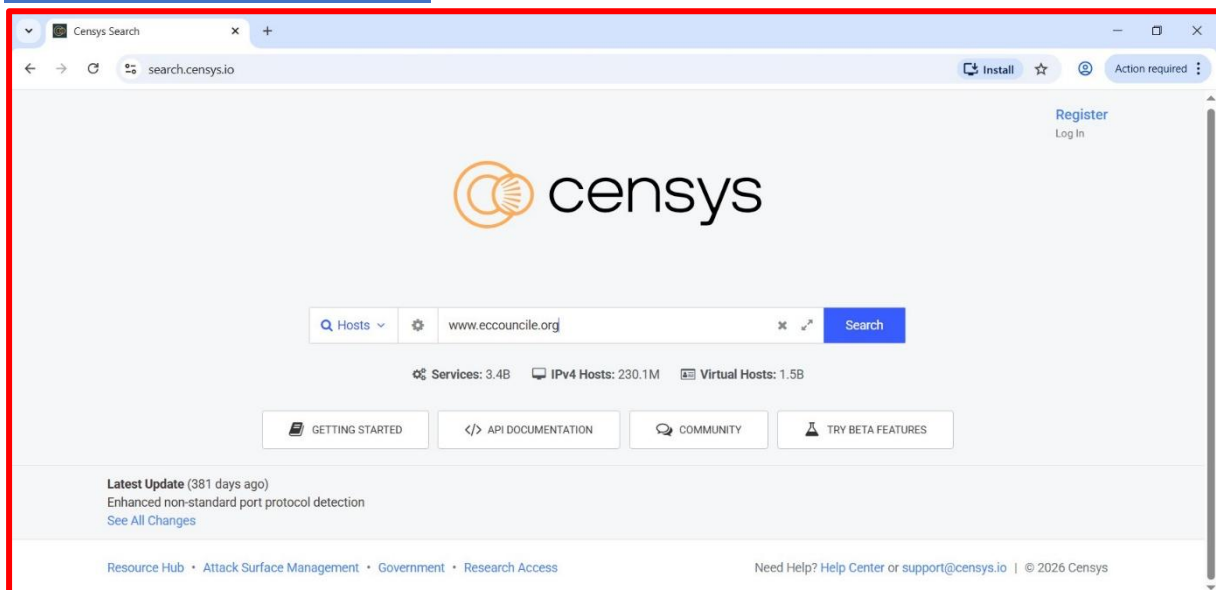
Identifying the operating system used by a target server can help attackers choose appropriate exploits.

In this task, we used the **Censys search platform** to discover OS details through passive reconnaissance.

Step 1: Open Censys

Open a browser and navigate to:

<https://search.censys.io>

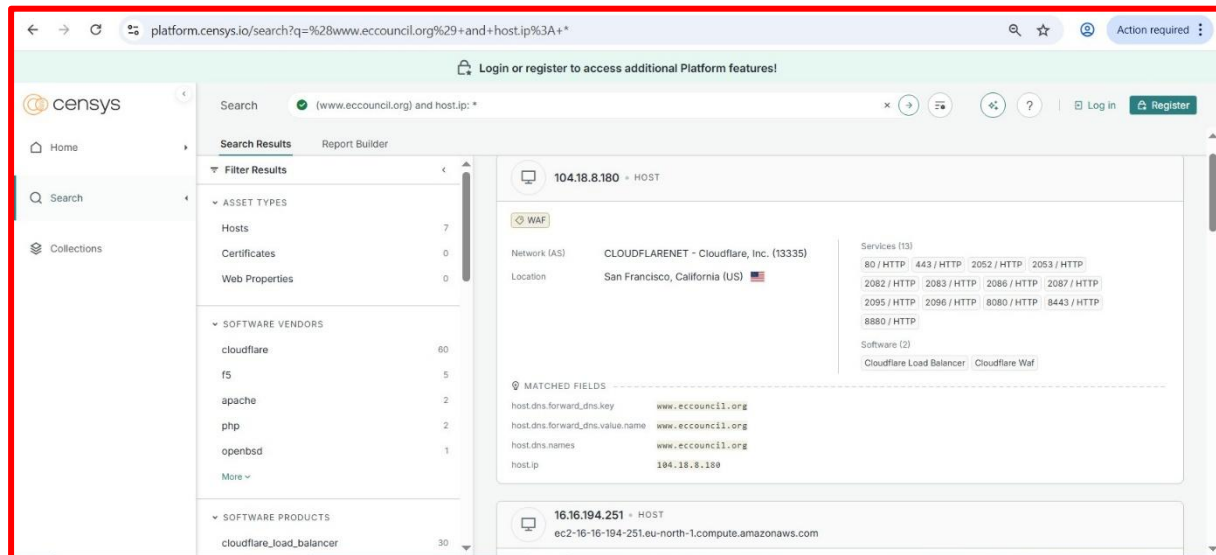


Step 2: Search Target Domain

Enter the target domain:

www.eccouncil.org

Press **Enter**.



Step 3: Select Host IP

Click any host IP address from the results.

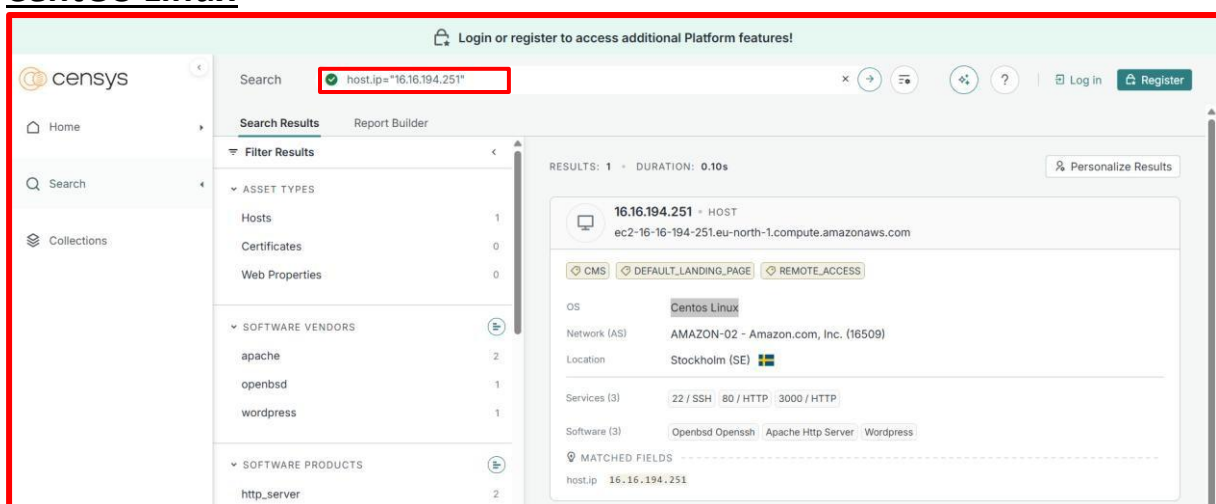
Step 4: Identify Operating System

The host information page displays several details including:

- *Operating system*
- *Open ports*
- *Location*
- *Network services*

Example OS discovered:

CentOS Linux



This information helps attackers identify vulnerabilities specific to the operating system.

4

Footprinting through Social Networking Sites

Footprinting through Social Networking Sites

Social networking platforms are widely used by individuals and organizations to communicate, share information, and build professional relationships. From a cybersecurity perspective, social networking sites are a rich source of **Open-Source Intelligence (OSINT)**. Common information that may be available on social networking sites includes:

- *Employee profiles*
- *Job roles and responsibilities*
- *Contact details*
- *Education history*
- *Organizational partnerships*
- *Company announcements*

Attackers may use this information to conduct targeted attacks such as **social engineering, phishing, and identity impersonation**.

Examples of popular social networking platforms include:

- **LinkedIn**
- **Facebook**
- **Instagram**
- **Twitter**
- **YouTube**

Task 01: Gathering Information using Sherlock

Sherlock is a **Python-based OSINT tool** designed to identify social media accounts associated with a specific username or individual.

The tool scans hundreds of social networking platforms and attempts to locate profiles related to the target.

Sherlock can help investigators discover:

- **Social media accounts**
- **Personal profiles**
- **Usernames across multiple platforms**
- **Links to public profiles**

Step 1: Start Parrot Security OS

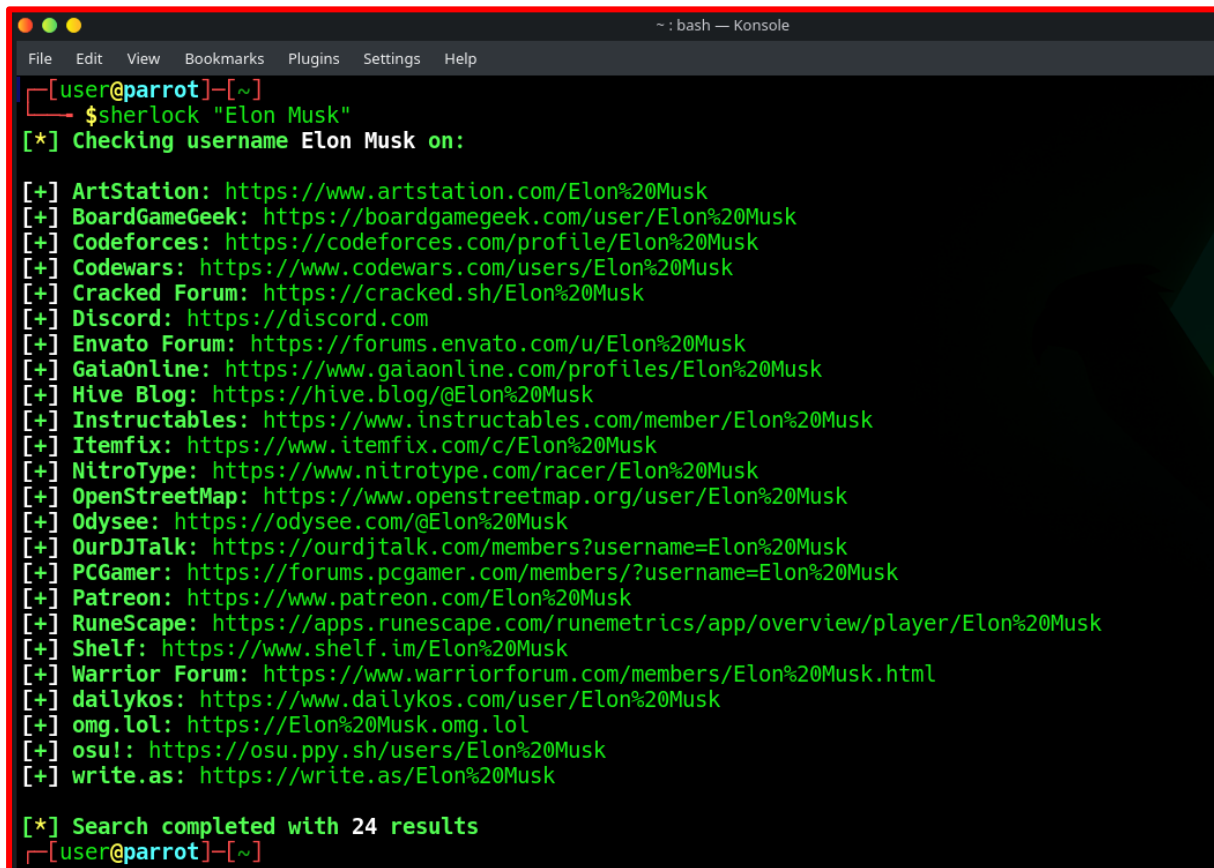
Turn on the **Parrot Security virtual machine**.

Open the **Terminal window**.

Step 2: Run Sherlock Command

In the terminal, execute the following command:

sherlock "Elon Musk"



```
~ : bash — Konsole
File Edit View Bookmarks Plugins Settings Help
[user@parrot]~
$ sherlock "Elon Musk"
[*] Checking username Elon Musk on:

[+] ArtStation: https://www.artstation.com/Elon%20Musk
[+] BoardGameGeek: https://boardgamegeek.com/user/Elon%20Musk
[+] Codeforces: https://codeforces.com/profile/Elon%20Musk
[+] Codewars: https://www.codewars.com/users/Elon%20Musk
[+] Cracked Forum: https://cracked.sh/Elon%20Musk
[+] Discord: https://discord.com
[+] Envato Forum: https://forums.envato.com/u/Elon%20Musk
[+] GaiaOnline: https://www.gaiaonline.com/profiles/Elon%20Musk
[+] Hive Blog: https://hive.blog/@Elon%20Musk
[+] Instructables: https://www.instructables.com/member/Elon%20Musk
[+] Itemfix: https://www.itemfix.com/c/Elon%20Musk
[+] NitroType: https://www.nitrotype.com/racer/Elon%20Musk
[+] OpenStreetMap: https://www.openstreetmap.org/user/Elon%20Musk
[+] Odysee: https://odysee.com/@Elon%20Musk
[+] OurDJTalk: https://ourdjtalk.com/members?username=Elon%20Musk
[+] PCGamer: https://forums.pcgamer.com/members/?username=Elon%20Musk
[+] Patreon: https://www.patreon.com/Elon%20Musk
[+] RuneScape: https://apps.runescape.com/runemetrics/app/overview/player/Elon%20Musk
[+] Shelf: https://www.shelf.im/Elon%20Musk
[+] Warrior Forum: https://www.warriorforum.com/members/Elon%20Musk.html
[+] dailykos: https://www.dailykos.com/user/Elon%20Musk
[+] omg.lol: https://Elon%20Musk.omg.lol
[+] osu!: https://osu.ppy.sh/users/Elon%20Musk
[+] write.as: https://write.as/Elon%20Musk

[*] Search completed with 24 results
[user@parrot]~
```

Step 3: Analyze Results

After the scan completes, the tool will display a list of URLs related to the target.

These URLs may include profiles from platforms such as:

- Twitter
- Instagram
- GitHub
- LinkedIn
- Reddit

5

Website Footprinting

Website Footprinting

Website footprinting involves gathering technical and administrative information about a target website. This information helps security analysts understand the structure and configuration of a web server. Information that can be obtained through website footprinting includes:

- *Domain ownership details*
- *IP address of the web server*
- *Hosting provider*
- *DNS records*
- *Server operating system*
- *Network infrastructure*

Task 01: Gathering Website Information using Ping Utility

Ping is a network diagnostic tool used to test whether a host is reachable over an IP network.

The command sends **ICMP Echo Request packets** to the target system and waits for a response.

Ping helps determine:

- *IP address of the target host*
- *Network latency*
- *Packet loss*
- *Connectivity status*

Step 1: Open Command Prompt

Turn on the **Windows 10 virtual machine**.

Open **Command Prompt** by typing:

cmd

in the Windows search bar.

Step 2: Execute Ping Command

Enter the following command:

ping www.tryhackme.com

```
C:\Users\HP>ping www.tryhackme.com

Pinging www.tryhackme.com [104.20.29.66] with 32 bytes of data:
Reply from 104.20.29.66: bytes=32 time=31ms TTL=55
Reply from 104.20.29.66: bytes=32 time=30ms TTL=55
Reply from 104.20.29.66: bytes=32 time=31ms TTL=55
Reply from 104.20.29.66: bytes=32 time=31ms TTL=55

Ping statistics for 104.20.29.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 30ms, Maximum = 31ms, Average = 30ms

C:\Users\HP>
```

The result displays the IP address of the target domain along with response times.

Example information obtained:

- *Target IP address*
- *Packets sent and received*
- *Round-trip time statistics*

Step 3: Test Maximum Packet Size

Next, run the following command:

ping www.tryhackme.com -f -l 1500

[Insert Screenshot: Ping Packet Fragmentation Result]

The -f option prevents packet fragmentation, while -l specifies packet size

```
C:\Users\HP>ping www.tryhackme.com -f -l 1500

Pinging www.tryhackme.com [172.66.164.239] with 1500 bytes of data:
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.

Ping statistics for 172.66.164.239:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\HP>
```


Step 4: Adjust Packet Size

Run another command:

ping www.tryhackme.com -f -l 1300

```
C:\Users\HP>ping www.tryhackme.com -f -l 1300

Pinging www.tryhackme.com [172.66.164.239] with 1300 bytes of data:
Reply from 172.66.164.239: bytes=1300 time=200ms TTL=48
Reply from 172.66.164.239: bytes=1300 time=265ms TTL=48
Reply from 172.66.164.239: bytes=1300 time=199ms TTL=48
Reply from 172.66.164.239: bytes=1300 time=205ms TTL=48

Ping statistics for 172.66.164.239:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 199ms, Maximum = 265ms, Average = 217ms

C:\Users\HP>
```

Task 02: Investigating Website using CentralOps

CentralOps is an online network investigation tool that provides information about domains, IP addresses, and DNS records.

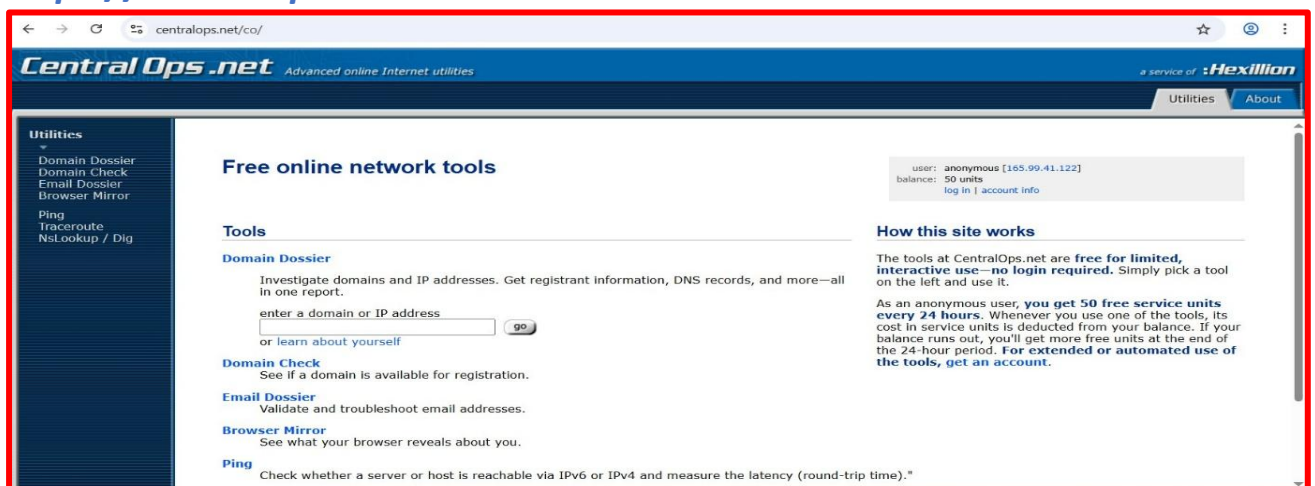
The platform offers services such as:

- ***Whois lookup***
- ***DNS records lookup***
- ***Traceroute analysis***
- ***Network information***

Step 1: Open CentralOps

Navigate to:

<https://centralops.net>

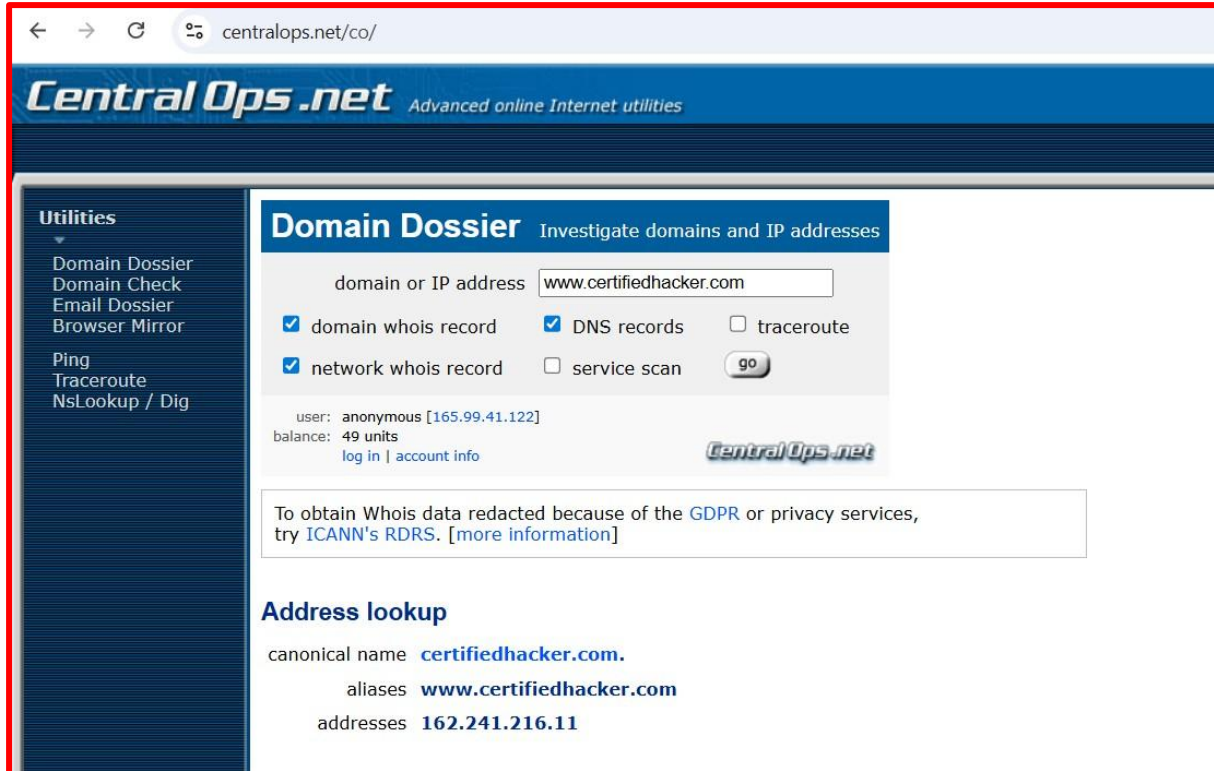


Step 2: Search Target Domain

Enter the following domain:

www.certifiedhacker.com

Click **Go**.



The screenshot shows the CentralOps.net website interface. The browser address bar displays 'centralops.net/co/'. The page header includes the 'CentralOps.net' logo and the tagline 'Advanced online Internet utilities'. On the left, a 'Utilities' sidebar lists options like 'Domain Dossier', 'Domain Check', 'Email Dossier', 'Browser Mirror', 'Ping', 'Traceroute', and 'NsLookup / Dig'. The main content area is titled 'Domain Dossier' with the subtitle 'Investigate domains and IP addresses'. It features a search input field containing 'www.certifiedhacker.com'. Below the input, there are checkboxes for 'domain whois record' (checked), 'DNS records' (checked), 'tracert' (unchecked), 'network whois record' (checked), and 'service scan' (unchecked). A 'go' button is positioned to the right of these options. Below the search options, user information is displayed: 'user: anonymous [165.99.41.122]' and 'balance: 49 units', with links for 'log in' and 'account info'. A message box states: 'To obtain Whois data redacted because of the GDPR or privacy services, try ICANN's RDRS. [more information]'. The 'Address lookup' section shows the canonical name as 'certifiedhacker.com.', aliases as 'www.certifiedhacker.com', and addresses as '162.241.216.11'.

Step 3: Review Domain Information

CentralOps displays various details including:

- *Domain Whois record*
- *DNS records*
- *Network Whois data*
- *IP address information*

Task 03: Generating Wordlist using CeWL

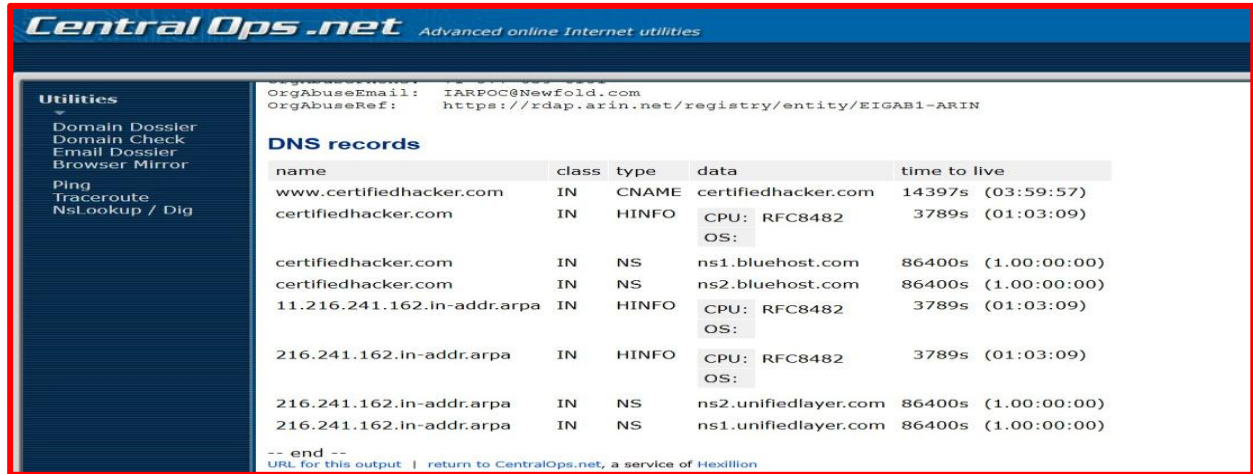
CeWL (Custom Word List Generator) is a tool that scans a website and extracts words that can be used for password cracking attacks. It works by crawling the target website and collecting unique words from its content.

These words may include:

- *Company terminology*
- *Product names*

- **Technical keywords**
- **Employee names**

Attackers can use these wordlists in **brute-force password attacks**.



The screenshot shows the CentralOps.net website with a sidebar menu on the left containing 'Utilities' and various tools like 'Domain Dossier', 'Domain Check', 'Email Dossier', 'Browser Mirror', 'Ping', 'Traceroute', and 'Nslookup / Dig'. The main content area displays DNS records for 'certifiedhacker.com'. At the top, it shows 'OrgAbuseEmail: IARPOC@Newfold.com' and 'OrgAbuseRef: https://rdap.arin.net/registry/entity/EIGAB1-ARIN'. Below this is a table of DNS records.

name	class	type	data	time to live
www.certifiedhacker.com	IN	CNAME	certifiedhacker.com	14397s (03:59:57)
certifiedhacker.com	IN	HINFO	CPU: RFC8482 OS:	3789s (01:03:09)
certifiedhacker.com	IN	NS	ns1.bluehost.com	86400s (1.00:00:00)
certifiedhacker.com	IN	NS	ns2.bluehost.com	86400s (1.00:00:00)
11.216.241.162.in-addr.arpa	IN	HINFO	CPU: RFC8482 OS:	3789s (01:03:09)
216.241.162.in-addr.arpa	IN	HINFO	CPU: RFC8482 OS:	3789s (01:03:09)
216.241.162.in-addr.arpa	IN	NS	ns2.unifiedlayer.com	86400s (1.00:00:00)
216.241.162.in-addr.arpa	IN	NS	ns1.unifiedlayer.com	86400s (1.00:00:00)

At the bottom, it says '-- end --' and 'URL for this output | return to CentralOps.net, a service of Hexillion'.

Step 1: Open Parrot Security Terminal

Switch to the **Parrot Security** virtual machine.

Open the **Terminal** window.

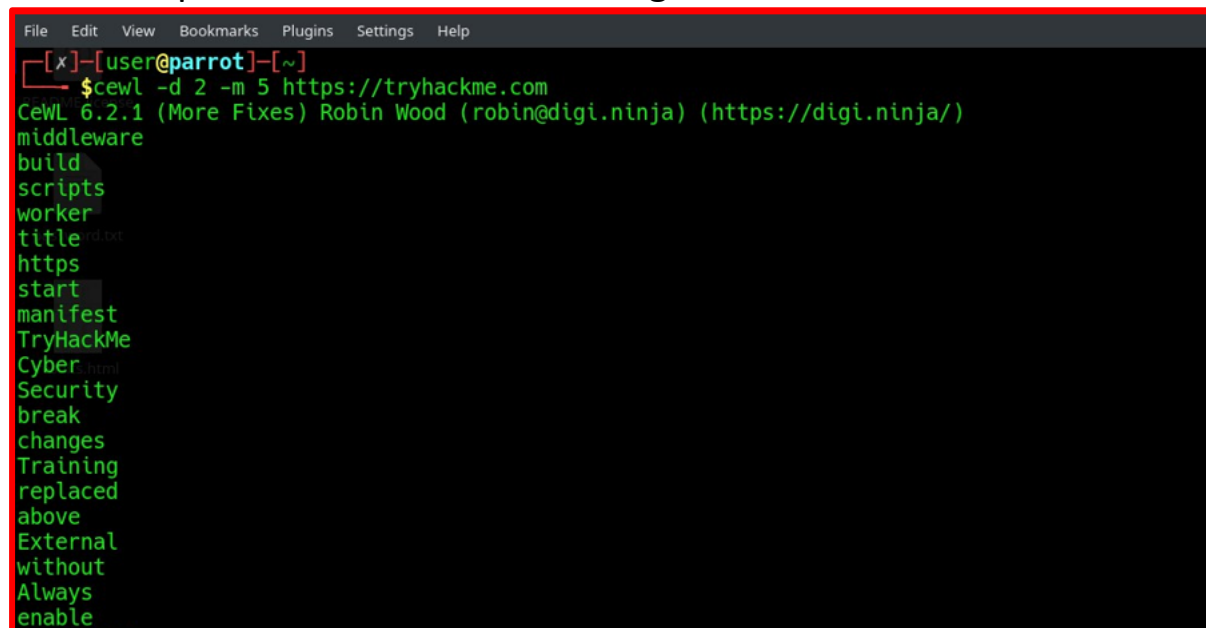
Step 2: Run CeWL Tool

Execute the following command:

[cewl -d 2 -m 5 https://tryhackme.com](https://tryhackme.com)

Where:

- **-d** specifies crawling depth
- **-m** specifies minimum word length



The screenshot shows a terminal window in a Parrot Security VM. The prompt is '[x]-[user@parrot]-[~]'. The command '\$cewl -d 2 -m 5 https://tryhackme.com' has been executed. The output shows the CeWL version (6.2.1) and the user (Robin Wood) along with a list of words extracted from the website: middleware, build, scripts, worker, title, https, start, manifest, TryHackMe, Cyber, Security, break, changes, Training, replaced, above, External, without, Always, enable.

Step 3: Save Wordlist to File

To save the generated wordlist, run:

[cewl -w wordlist.txt -d 2 -m 5 https://tryhackme.com](#)

```
fundamentals
Graph
Twitter
other
added
Cloudflare
Worker
includes
dynamic
based
pages
replace
correct
github
tryhackme
cloudflare
workers
master
Please
[user@parrot]~$ cewl -w wordlist.txt -d 2 -m 5 https://tryhackme.com
```

Step 4: View Wordlist

To display the saved file, run:

[cat wordlist.txt](#)

```
File Edit View Bookmarks Plugins Settings Help
[user@parrot]~$ cat wordlist.txt
middleware
build
scripts
worker
title
https
start
manifest
TryHackMe
Cyber
Security
break
changes
Training
replaced
```

The extracted wordlist may later be used in password cracking tools such as:

- [Hydra](#)
- [John the Ripper](#)
- [Hashcat](#)

6

Whois Footprinting

Whois footprinting is a technique used to collect registration details about a domain name or IP address.

. These records may contain important information such as:

- *Domain registration date*
- *Expiration date*
- *Domain registrar*
- *Name servers*
- *Organization name*
- *Contact information*

6.1 Overview of Whois

Whois is a query protocol used to retrieve information stored in databases that maintain records of internet resources. These resources may include:

- *Domain names*
- *IP address ranges*
- *Autonomous system numbers*

Whois servers operate over **TCP port 43** and respond to requests by providing textual information about the queried resource.

Regional Internet Registries (RIRs) maintain these databases. Some of the major RIRs include:

- **ARIN (North America)**
- **RIPE NCC (Europe)**
- **APNIC (Asia-Pacific)**
- **AFRINIC (Africa)**
- **LACNIC (Latin America)**

The information returned by Whois queries helps security professionals identify the ownership and administrative details of a domain.

Task 01: Performing Whois Lookup using DomainTools

In this lab exercise, the **DomainTools Whois lookup service** was used to gather domain registration details.

Step 1: Open DomainTools

Turn on the **Windows virtual machine** and launch a web browser.

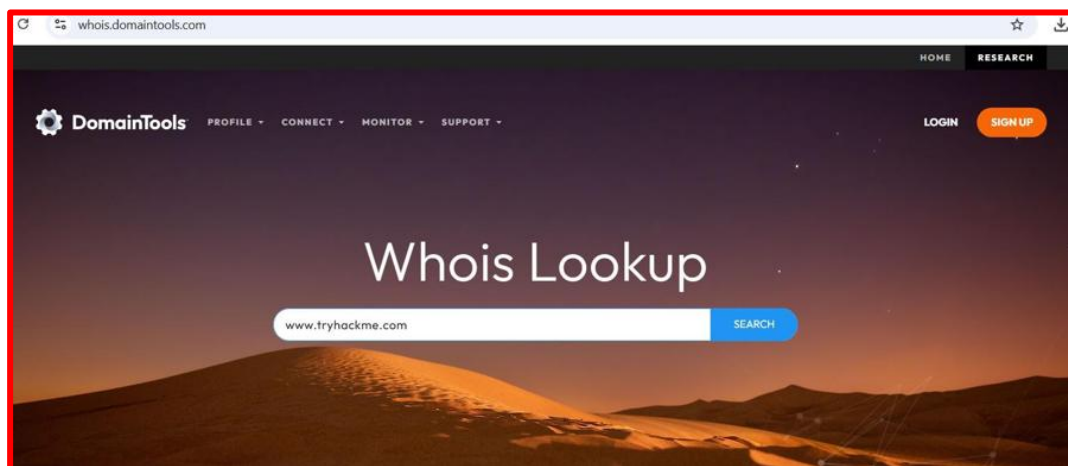
Navigate to the following website:

<http://whois.domaintools.com>

Step 2: Search Target Domain

Enter the target domain name in the search bar.

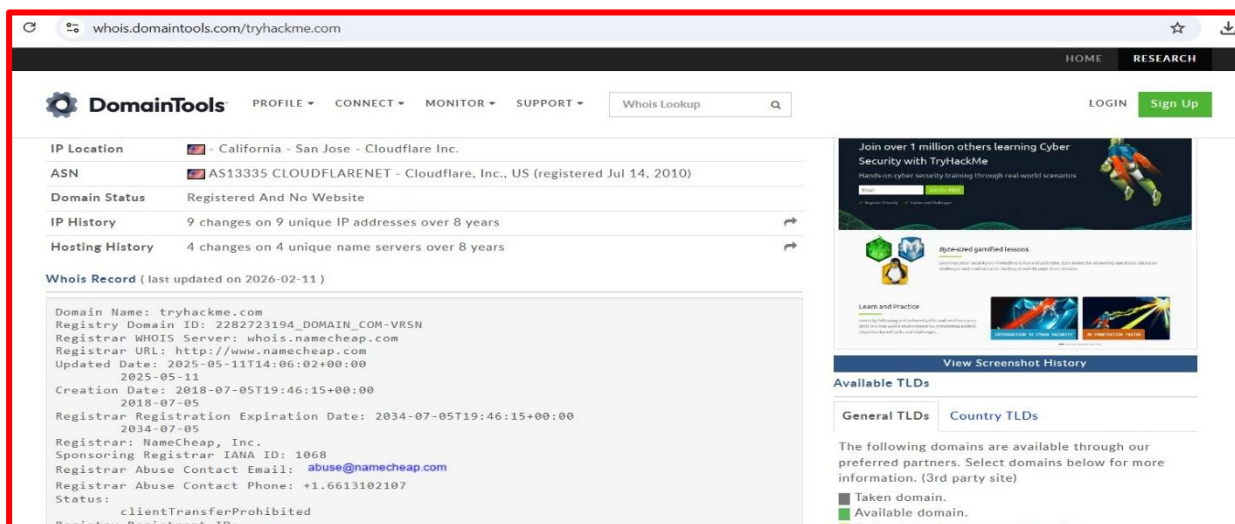
Click **Search**.



Step 3: Analyze Whois Results

The Whois lookup results display various pieces of information such as:

- *Registrar name*
- *Creation and expiration dates*
- *Name servers*
- *IP address information*
- *Domain status*



7

DNS Footprinting

DNS Footprinting

DNS footprinting is the process of gathering information related to a domain's DNS records. **The Domain Name System (DNS)** is responsible for translating human-readable domain names into machine-readable IP addresses.

The Domain Name System acts as an intermediary between users and internet services by translating domain names into IP addresses.

For example:

www.tryhackme.com → 104.20.29.66.

DNS servers maintain records that describe how different services of a domain are structured. These records can reveal valuable information about the target network.

Task 01: Gathering DNS Information using nslookup

The **nslookup** command-line tool is commonly used to query DNS servers and retrieve domain information.

Step 1: Launch Command Prompt

Turn on the **Windows 10 virtual machine**.

Open **Command Prompt** by typing:

Cmd

Step 2: Start nslookup

In the command prompt, type:

nslookup

The system will display the default DNS server assigned to the machine.


```
Administrator: Command Prompt - nslookup
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nslookup
DNS request timed out.
    timeout was 2 seconds.
Default Server:  UnKnown
Address:  192.168.47.2

> set type=a
> www.tryhackme.com
Server:  UnKnown
Address:  192.168.47.2

Name:    www.tryhackme.com.localdomain
Address:  104.20.29.66
```

Step 3: Query IP Address Record

Enter the following command:

[set type=a](#)

Then type the target domain:

www.tryhackme.com

```
Administrator: Command Prompt - nslookup
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nslookup
DNS request timed out.
    timeout was 2 seconds.
Default Server:  UnKnown
Address:  192.168.47.2

> set type=a
> www.tryhackme.com
Server:  UnKnown
Address:  192.168.47.2

Name:    www.tryhackme.com.localdomain
Address:  104.20.29.66

> set type=cname
> tryhackme.com
Server:  UnKnown
Address:  192.168.47.2
```

This query retrieves the IP address associated with the domain.

Step 4: Query CNAME Records

Next, change the query type to CNAME:

set type=cname

Then enter:

www.tryhackme.com

```
> set type=cname
> tryhackme.com
Server: UnKnown
Address: 192.168.47.2

DNS request timed out.
  timeout was 2 seconds.
tryhackme.com
  primary name server = kip.ns.cloudflare.com
  responsible mail addr = dns.cloudflare.com
  serial      = 2395288233
  refresh    = 10000 (2 hours 46 mins 40 secs)
  retry      = 2400 (40 mins)
  expire     = 604800 (7 days)
  default TTL = 1800 (30 mins)

> set type=a
> kip.ns.cloudflare.com
Server: UnKnown
Address: 192.168.47.2

Name:   kip.ns.cloudflare.com.localdomain
Address: 172.64.33.128

> |
```

This command identifies alias records that point to the domain.

DNS Lookup using Online Tools

Open the following website:

<http://www.kloth.net/services/nslookup.php>

Enter the target domain:

www.tryhackme.com

Select **A (IPv4 address)** as the query type and click **Look it up**.

NSLOOKUP: look up and find IP addresses in the DNS

Query a DNS domain nameserver to lookup and find IP address information of computers in the internet. Convert a host or domain name to an IP address.

This is the right place for you to check how your web hosting company or domain name registrar has set up the DNS stuff for your domain, how your dynamic DNS service works, or how to prevent internet abuse.

This online service is for private non-commercial use only. Please do not abuse. No automated queries. No bots.

NSlookup

Domain:

... the name of the machine to look up.

Server:

... the DNS nameserver you want to handle your query (just start with this site's default server)

Query:

... here is the **nslookup** result for **www.tryhackme.com** from server localhost, querytype=A :

```
DNS server handling your query: localhost
DNS server's address: ::1#53

Non-authoritative answer:
Name:   www.tryhackme.com
Address: 104.20.29.66
Name:   www.tryhackme.com
Address: 172.66.164.239
```

8 Network Footprinting

Network Footprinting

Network footprinting involves collecting information about the structure and connectivity of a target network.

This process helps ethical hackers understand:

- *Network topology*
- *Firewall locations*
- *Network devices between source and destination*

Network footprinting tools trace the route packets travel between systems.

Task 01: Network Tracerouting using Windows

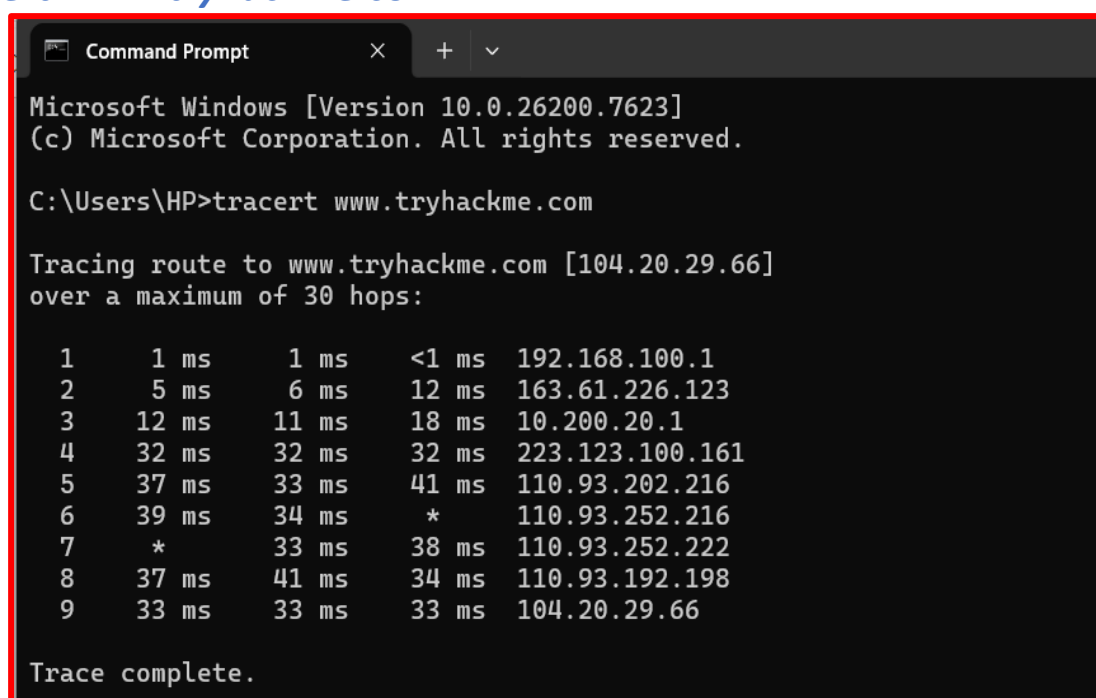
Step 1: Open Command Prompt

Open **Command Prompt** on the Windows system.

Step 2: Run Traceroute Command

Enter the following command:

tracert www.tryhackme.com



```
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>tracert www.tryhackme.com

Tracing route to www.tryhackme.com [104.20.29.66]
over a maximum of 30 hops:

  1    1 ms    1 ms    <1 ms   192.168.100.1
  2    5 ms    6 ms    12 ms   163.61.226.123
  3   12 ms   11 ms   18 ms   10.200.20.1
  4   32 ms   32 ms   32 ms   223.123.100.161
  5   37 ms   33 ms   41 ms   110.93.202.216
  6   39 ms   34 ms    *      110.93.252.216
  7    *      33 ms   38 ms   110.93.252.222
  8   37 ms   41 ms   34 ms   110.93.192.198
  9   33 ms   33 ms   33 ms   104.20.29.66

Trace complete.
```

Step 3: View Command Options

To view additional options, run:

[tracert /?](#)

Step 4: Limit Maximum Hops

To limit the number of hops, run:

[tracert -h 5 www.certifiedhacker.com](#)

Where:

- **-h** specifies the maximum number of hops allowed.

```
C:\Users\HP>tracert /?

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
  -d                Do not resolve addresses to hostnames.
  -h maximum_hops   Maximum number of hops to search for target.
  -j host-list       Loose source route along host-list (IPv4-only).
  -w timeout         Wait timeout milliseconds for each reply.
  -R                Trace round-trip path (IPv6-only).
  -S srcaddr         Source address to use (IPv6-only).
  -4                Force using IPv4.
  -6                Force using IPv6.

C:\Users\HP>tracert -h 5 www.tryhackme.com

Tracing route to www.tryhackme.com [172.66.164.239]
over a maximum of 5 hops:

  1    1 ms    1 ms    1 ms  192.168.100.1
  2    *        3 ms    3 ms  163.61.226.123
  3    *       13 ms   10 ms  10.200.20.1
  4   35 ms   33 ms   32 ms  223.123.100.161
  5   31 ms   31 ms   31 ms  10.180.161.66

Trace complete.

C:\Users\HP>
```

Task 02: Network Tracerouting using Linux

Traceroute can also be performed in Linux environments such as Parrot OS.

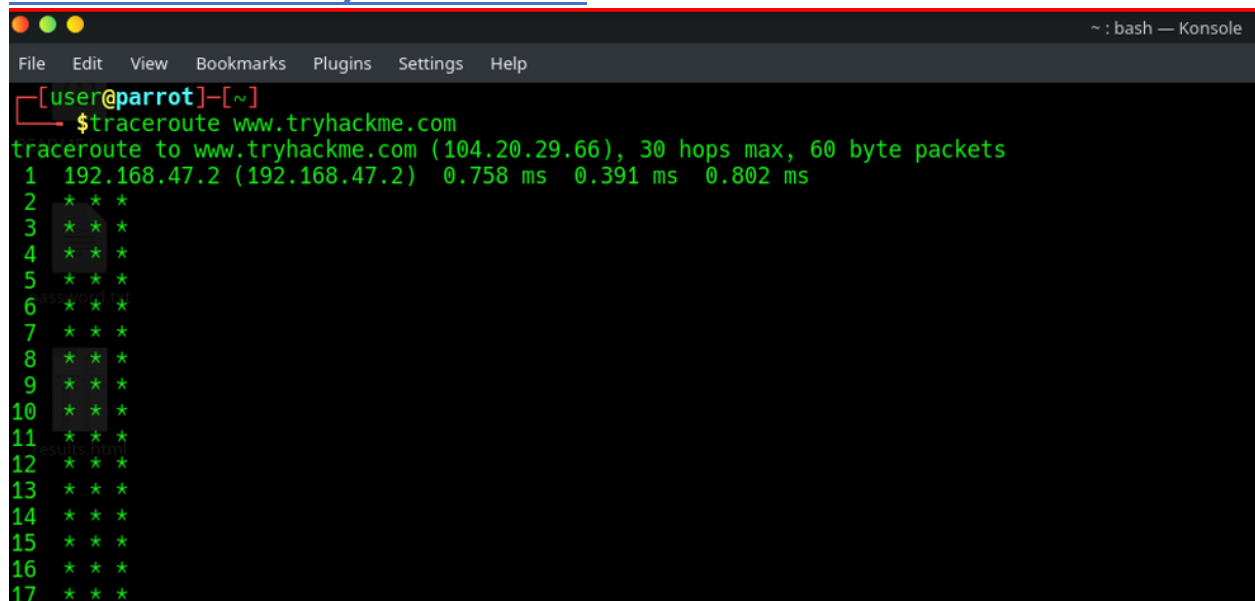
Step 1: Start Parrot Security OS

Open the **Parrot Security** virtual machine and launch the **Terminal**.

Step 2: Run traceroute Command

Execute the following command:

[tracert www.tryhackme.com](https://www.tryhackme.com)



```
[user@parrot]~  
$tracert www.tryhackme.com  
tracert to www.tryhackme.com (104.20.29.66), 30 hops max, 60 byte packets  
 1  192.168.47.2 (192.168.47.2)  0.758 ms  0.391 ms  0.802 ms  
 2  * * *  
 3  * * *  
 4  * * *  
 5  * * *  
 6  * * *  
 7  * * *  
 8  * * *  
 9  * * *  
10  * * *  
11  * * *  
12  * * *  
13  * * *  
14  * * *  
15  * * *  
16  * * *  
17  * * *
```

The result displays each intermediate system that packets pass through before reaching the destination.

9 Email Footprinting

Email Footprinting

Email footprinting is the process of analyzing email messages in order to collect information about the sender and the route the email traveled through before reaching the recipient. Every email contains a **header section** that includes technical details about the message.

These headers store important information such as:

- *Sender's IP address*
- *Mail server details*
- *Email routing path*
- *Timestamp information*
- *Authentication details*

9.1 Overview of Email Footprinting

Email footprinting involves tracking the origin and path of an email message. When an email is sent, it travels through several mail servers

before reaching its final destination. Each server adds information to the email header.

This header data can reveal important details such as:

- *Sender's IP address*
- *Mail server location*
- *Operating system used by the sender*
- *Email transmission time*
- *Geographic location of the sender*

Task 01: Email Header Analysis using eMailTrackerPro

In this task, the **eMailTrackerPro** tool was used to analyze email headers and trace the path of an email message.

Step 1: Install eMailTrackerPro

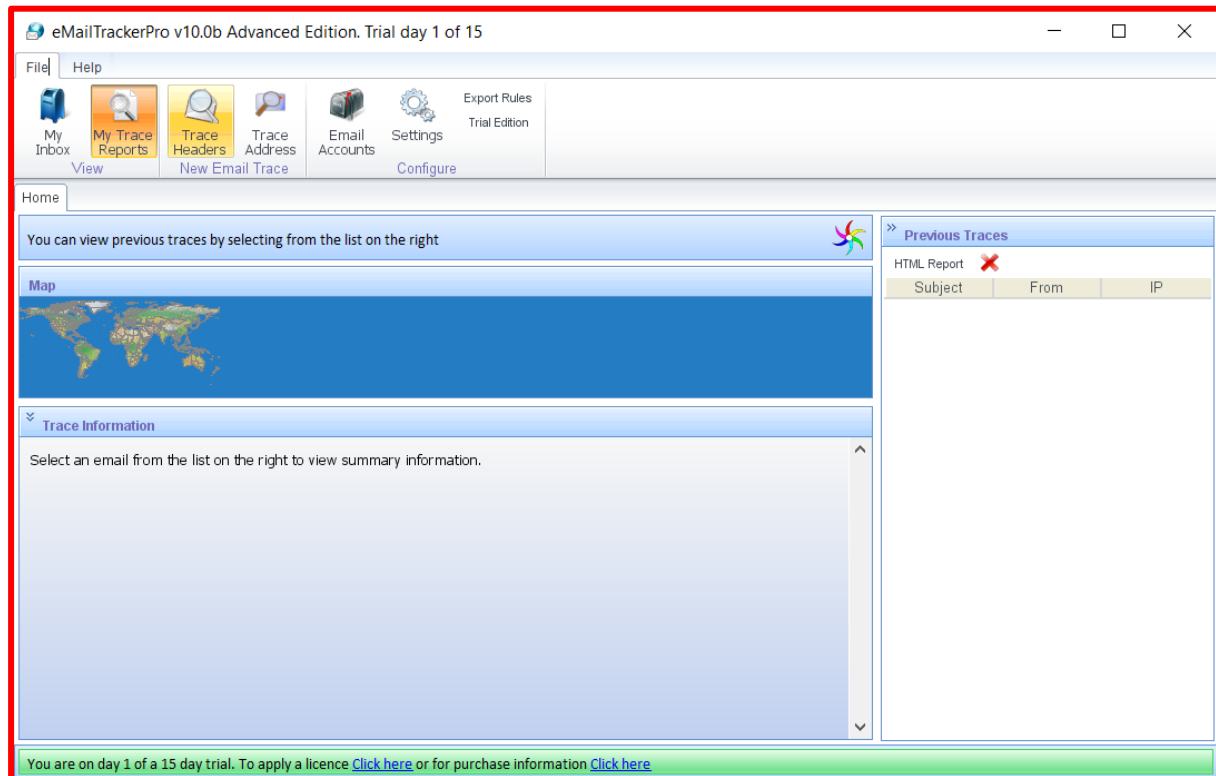
Turn on the **Windows virtual machine**.

Navigate to the folder containing the **eMailTrackerPro** installation file (**.exe**) and double-click it.

Step 2: Launch the Application

After installation, launch **eMailTrackerPro**.

The main dashboard of the application will appear.



Step 3: Start Email Trace

Click on the **My Trace Reports** section.

Then select:

Trace Headers

I Want To: _____

☒ **Trace an email I have received**

A received email message often contains information that can locate the computer where the message was composed, the company name and sender's ISP ([more info](#)).

☐ **Look up network responsible for an email address**

An email address lookup will find information about the network responsible for mail sent from that address. It will not get any information about the sender of mail from an address but can still produce useful information.

Enter Details _____

To proceed, paste the email headers in the box below ([how do I find the headers?](#)).

Note: If you are using Microsoft Outlook, you can trace an email message directly from Outlook by using the eMailTrackerPro shortcut on the toolbar.

Email headers:

Trace **Cancel**

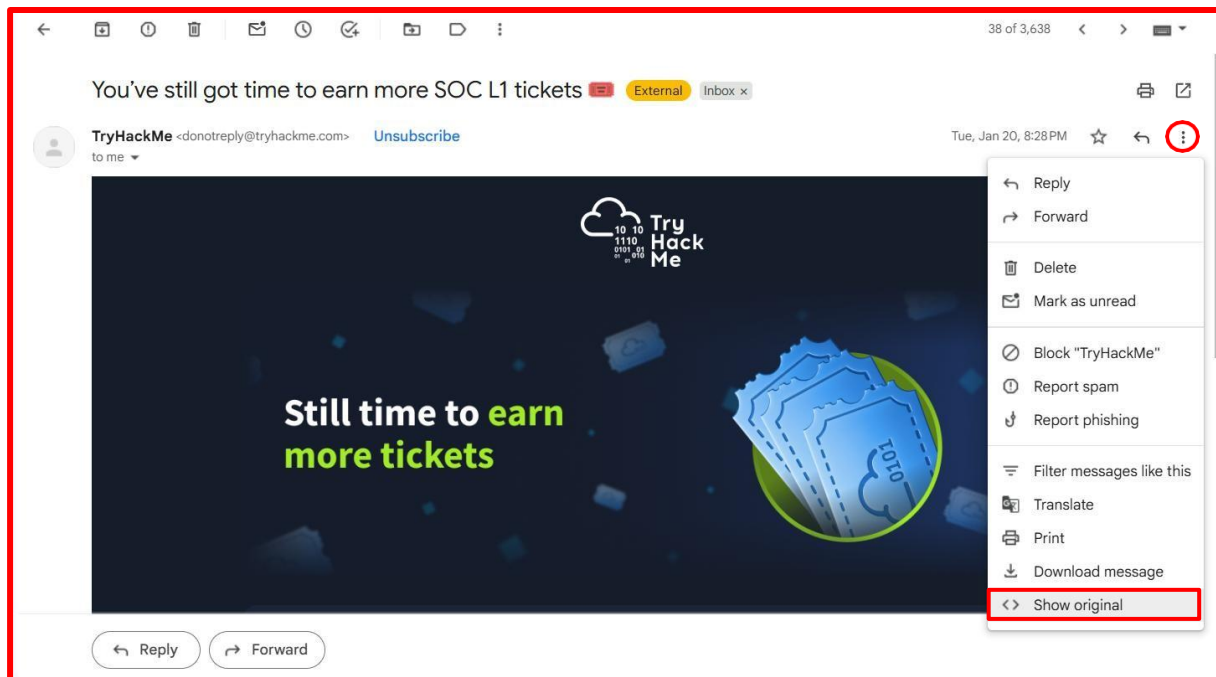
Step 4: Obtain Email Header from Gmail

To analyze an email header, open your Gmail inbox.

Follow these steps:

1. Open any email message

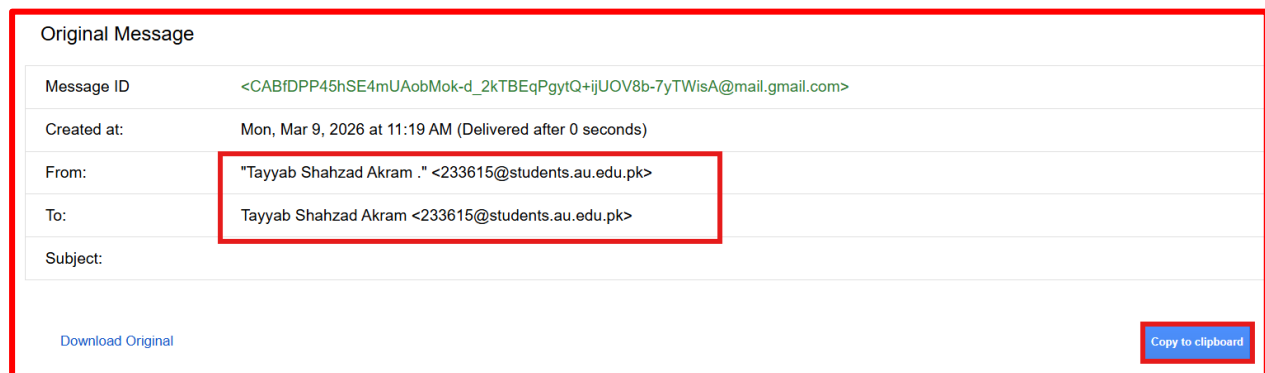
2. Click the **three-dot menu (More options)** beside the reply button
3. Select **Show Original**



Step 5: Copy Email Header

The full email header will appear in a new tab.

Click **Copy to Clipboard**.



Step 6: Paste Header into eMailTrackerPro

Return to **eMailTrackerPro** and paste the copied header into the **Email Headers** field.

Click **Trace**.

Enter Details

To proceed, paste the email headers in the box below ([how do I find the headers?](#)).

Note: If you are using Microsoft Outlook, you can trace an email message directly from Outlook by using the eMailTrackerPro shortcut on the toolbar.

Email headers:

```
Delivered-To: 233610@students.au.edu.pk
Received: Delivered-To: 233610@students.au.edu.pk
Received: by 2002:a05:7022:6184:b0:10e:21f4:a6b5 with SMTP id n4csp3
      Tue, 20 Jan 2026 07:28:23 -0800 (PST)
X-Received: by 2002:ac8:5dd1:0:b0:4f1:e989:4592 with SMTP id d75a77k
      Tue, 20 Jan 2026 07:28:23 -0800 (PST)
ARC-Seal: i=1; a=rsa-sha256; t=1768922903; cv=none;
      d=google.com; s=arc-20240605;
```

Trace **Cancel**

Step 7: Analyze Trace Results

The tool will generate a report showing:

- Email route
- IP addresses of intermediate servers
- Geographic location of the sender
- Network details of mail servers

The tool also displays the route on a **world map** interface.

The screenshot shows the eMailTrackerPro interface. On the left, there's a world map with a red line indicating the email route. Below the map is a table with columns: Hop IP, Hop Name, and Location.

Hop IP	Hop Name	Location
192.168.47.2		
159.112.241.138	v6138 v5c4dda55 use4.send.mailgun.net	San Antonio, TX

On the right, there's a 'Network Whois' panel with the following details:

```
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# if you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange: 159.112.240.0 - 159.112.255.255
CIDR: 159.112.240.0/20
NetName: MT-757
NetHandle: NET-159-112-240-0-1
Parent: NET159 (NET-159-0-0-0-0)
NetType: Direct Allocation
OriginAS:
Organization: Mailgun Technologies Inc. (MT-757)
RegDate: 2022-02-10
Updated: 2022-07-11
Comment:
Comment: —BEGIN CERTIFICATE—MIID5jCCAJCCQC19dNdmG0nzANE
Ref: https://rdap.arin.net/registry/ip/159.112.240.0

OrgName: Mailgun Technologies Inc.
OrgId: MT-757
```

Below the Network Whois panel is a 'Domain Whois' section.

Footprinting using Various Tools

Footprinting using Various Tools

Modern penetration testing relies heavily on automated tools to collect information efficiently. Some of the commonly collected information includes:

- *IP addresses*
- *Domain information*
- *DNS records*
- *Email addresses*
- *Employee information*

One of the powerful tools used for reconnaissance is **Recon-ng**.

Step 1: Start Parrot Security OS

Turn on the **Parrot Security virtual machine** and open the **Terminal**.

Step 2: Launch Recon-ng

Run the following command:

recon-ng

[illegible]

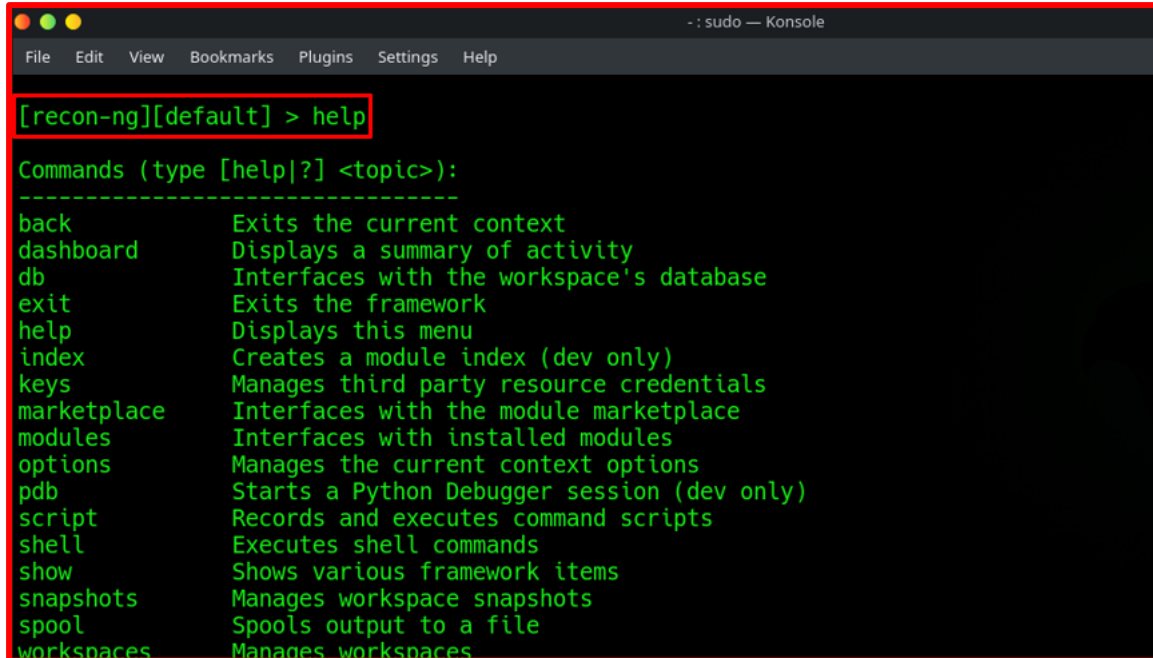
The Recon-ng console environment will open.

Step 3: View Available Commands

Type:

[help](#)

This command displays all available commands and their functions.



```
-- : sudo — Konsole
File Edit View Bookmarks Plugins Settings Help

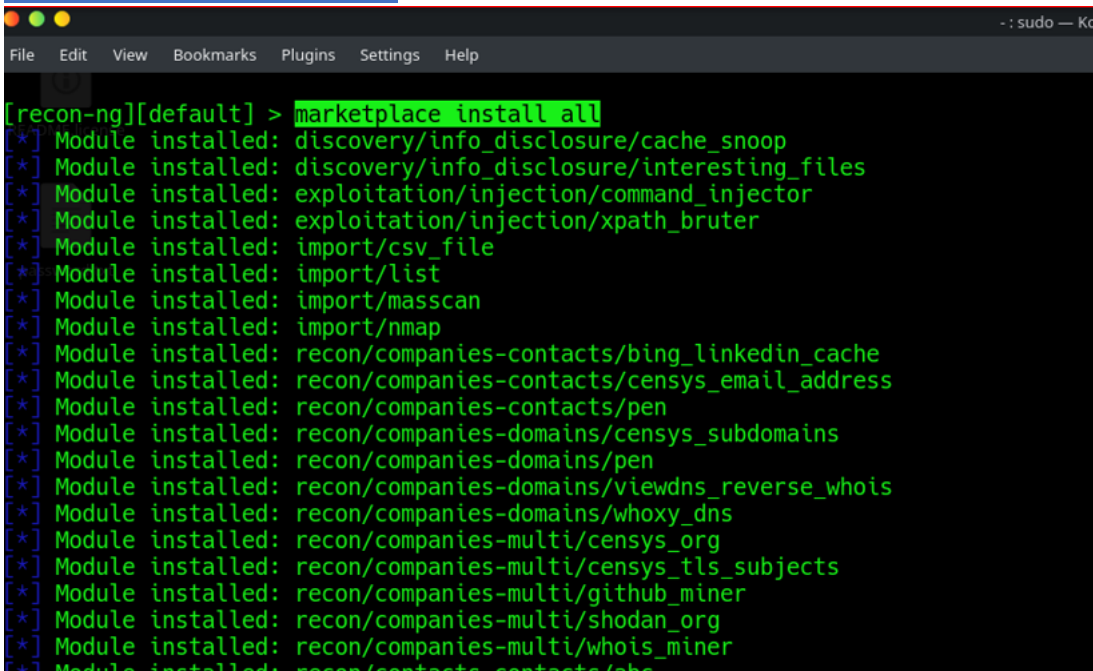
[recon-ng][default] > help

Commands (type [help|?] <topic>):
-----
back           Exits the current context
dashboard      Displays a summary of activity
db             Interfaces with the workspace's database
exit           Exits the framework
help           Displays this menu
index          Creates a module index (dev only)
keys           Manages third party resource credentials
marketplace    Interfaces with the module marketplace
modules        Interfaces with installed modules
options        Manages the current context options
pdb            Starts a Python Debugger session (dev only)
script         Records and executes command scripts
shell          Executes shell commands
show           Shows various framework items
snapshots      Manages workspace snapshots
spool          Spools output to a file
workspaces     Manages workspaces
```

Step 4: Install Modules

To install all modules available in the Recon-ng marketplace, run:

[marketplace install all](#)



```
-- : sudo — Konsole
File Edit View Bookmarks Plugins Settings Help

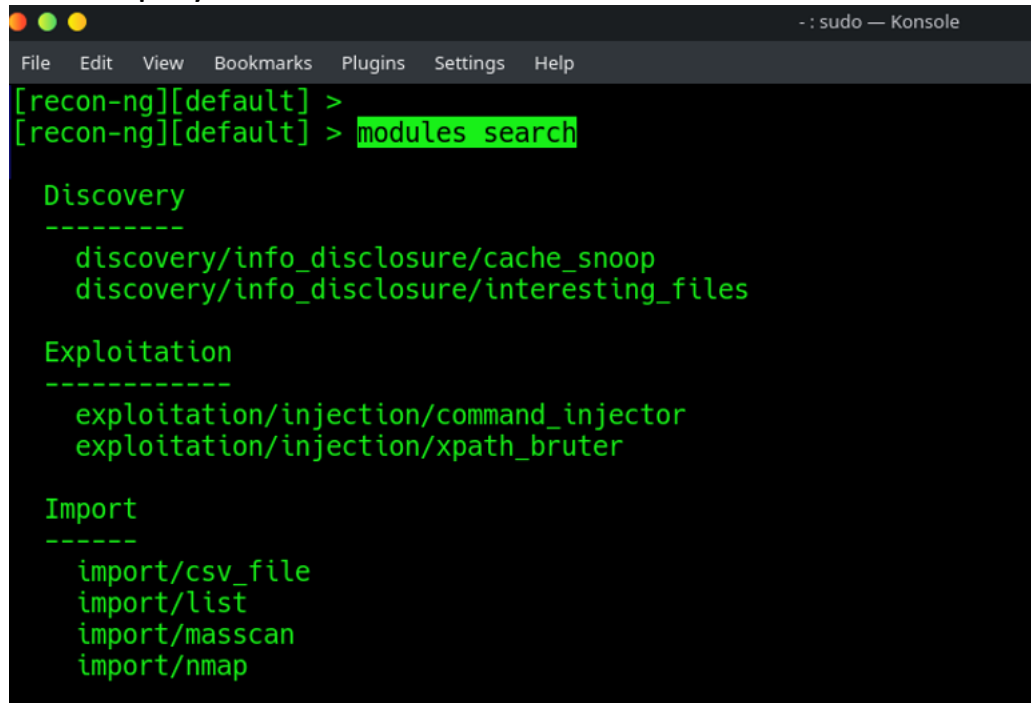
[recon-ng][default] > marketplace install all
[*] Module installed: discovery/info_disclosure/cache_snoop
[*] Module installed: discovery/info_disclosure/interesting_files
[*] Module installed: exploitation/injection/command_injector
[*] Module installed: exploitation/injection/xpath_bruter
[*] Module installed: import/csv_file
[*] Module installed: import/list
[*] Module installed: import/masscan
[*] Module installed: import/nmap
[*] Module installed: recon/companies-contacts/bing_linkedin_cache
[*] Module installed: recon/companies-contacts/censys_email_address
[*] Module installed: recon/companies-contacts/pen
[*] Module installed: recon/companies-domains/censys_subdomains
[*] Module installed: recon/companies-domains/pen
[*] Module installed: recon/companies-domains/viewdns_reverse_whois
[*] Module installed: recon/companies-domains/whoxy_dns
[*] Module installed: recon/companies-multi/censys_org
[*] Module installed: recon/companies-multi/censys_tls_subjects
[*] Module installed: recon/companies-multi/github_miner
[*] Module installed: recon/companies-multi/shodan_org
[*] Module installed: recon/companies-multi/whois_miner
[*] Module installed: recon/contacts-contacts/abc
```

Step 5: Search Available Modules

Run the command:

[modules search](#)

This displays all modules available for reconnaissance activities.



```
-- sudo — Konsole
File Edit View Bookmarks Plugins Settings Help
[recon-ng][default] >
[recon-ng][default] > modules search

Discovery
-----
discovery/info_disclosure/cache_snoop
discovery/info_disclosure/interesting_files

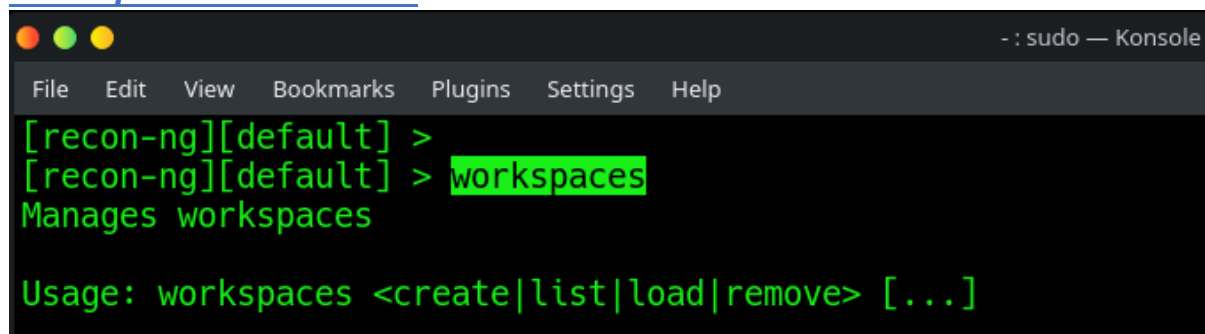
Exploitation
-----
exploitation/injection/command_injector
exploitation/injection/xpath_bruter

Import
-----
import/csv_file
import/list
import/masscan
import/nmap
```

Step 6: Create Workspace

Create a workspace named **CEH** using:

[workspaces create CEH](#)



```
-- sudo — Konsole
File Edit View Bookmarks Plugins Settings Help
[recon-ng][default] >
[recon-ng][default] > workspaces
Manages workspaces

Usage: workspaces <create|list|load|remove> [...]
```

Step 7: Add Target Domain

Insert the target domain into the database.

Run:

[db insert domains](#)

Enter:

tryhackme.com


```
[recon-ng][default] > workspaces create CEH
/root/.recon-ng/modules/recon/companies-contacts/bing_linkedin_cache.py:59: SyntaxWarning: invalid escape
sequence '\d'
  if fname is None or 'LinkedIn' in fullname or 'profiles' in name.lower() or re.search('^\\d+$',fname):
/root/.recon-ng/modules/recon/companies-contacts/bing_linkedin_cache.py:99: SyntaxWarning: invalid escape
sequence '\\.'
  m = re.search('\\.{3} (?P<title>.+?) at ', snippet)
[!] 'bing_api' key not set. bing_linkedin_cache module will likely fail at runtime. See 'keys add'.
[!] Module 'recon/companies-contacts/censys_email_address' disabled. Dependency required: 'censys'.
[!] Module 'recon/companies-domains/censys_subdomains' disabled. Dependency required: 'censys'.
[!] 'whoxy_api' key not set. whoxy_dns module will likely fail at runtime. See 'keys add'.
[!] Module 'recon/companies-multi/censys_org' disabled. Dependency required: 'censys'.
[!] Module 'recon/companies-multi/censys_tls_subjects' disabled. Dependency required: 'censys'.
[!] 'github_api' key not set. github_miner module will likely fail at runtime. See 'keys add'.
[!] Module 'recon/companies-multi/shodan_org' disabled. Dependency required: 'shodan'.
/root/.recon-ng/modules/recon/contacts-contacts/mangle.py:35: SyntaxWarning: invalid escape sequence '\\s'
  sub_pattern = '\\s'
```

Step 8: Discover Hosts

Load the module:

[modules load recon/domains-hosts/brute hosts](#)

Then run:

run

```
[recon-ng][CEH] > workspaces list

+-----+-----+
| Workspaces | Modified |
+-----+-----+
| CEH        | 2026-02-12 16:01:33 |
| default    | 2026-02-12 15:58:04 |
+-----+-----+

[recon-ng][CEH] > db insert domains
domain (TEXT): tryhackme.com
notes (TEXT):
[*] 1 rows affected.
[recon-ng][CEH] > show domains

+-----+-----+-----+-----+
| rowid | domain      | notes | module      |
+-----+-----+-----+-----+
| 1      | tryhackme.com |      | user_defined |
+-----+-----+-----+-----+

[*] 1 rows returned
```

Step 9: Perform Reverse Lookup

Load the reverse lookup module:

modules load recon/hosts-hosts/reverse_resolve

Then run:

```
-- : sudo — Konsole
File Edit View Bookmarks Plugins Settings Help

+-----+
| rowid | domain | notes | module |
+-----+
| 1 | tryhackme.com | | user_defined |
+-----+

[*] 1 rows returned
[recon-ng][CEH] > modules load brute
[*] Multiple modules match 'brute'.

Exploitation
-----
exploitation/injection/xpath_bruter

Recon
-----
recon/domains-domains/brute_suffix
recon/domains-hosts/brute_hosts
```

Step 10: Generate Recon Report

To generate a report, load the reporting module:

modules load reporting/html

Set report parameters:

options set FILENAME /home/user/Desktop/results.html

options set CREATOR Mudassir

options set CUSTOMER tryHackme Networks

Then execute:

run

```
-- : sudo — Konsole
File Edit View Bookmarks Plugins Settings Help

Recon
-----
recon/domains-domains/brute_suffix
recon/domains-hosts/brute_hosts

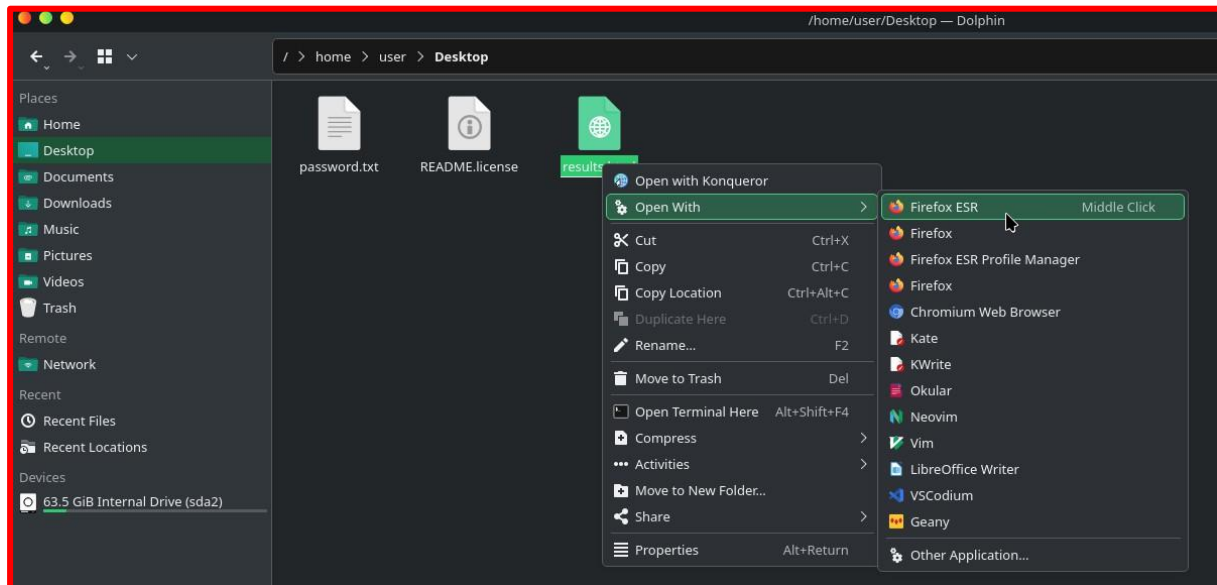
[recon-ng][CEH] > modules load recon/domains-hosts/brute_hosts
[recon-ng][CEH][brute_hosts] > run
password.txt

TRYHACKME.COM
-----

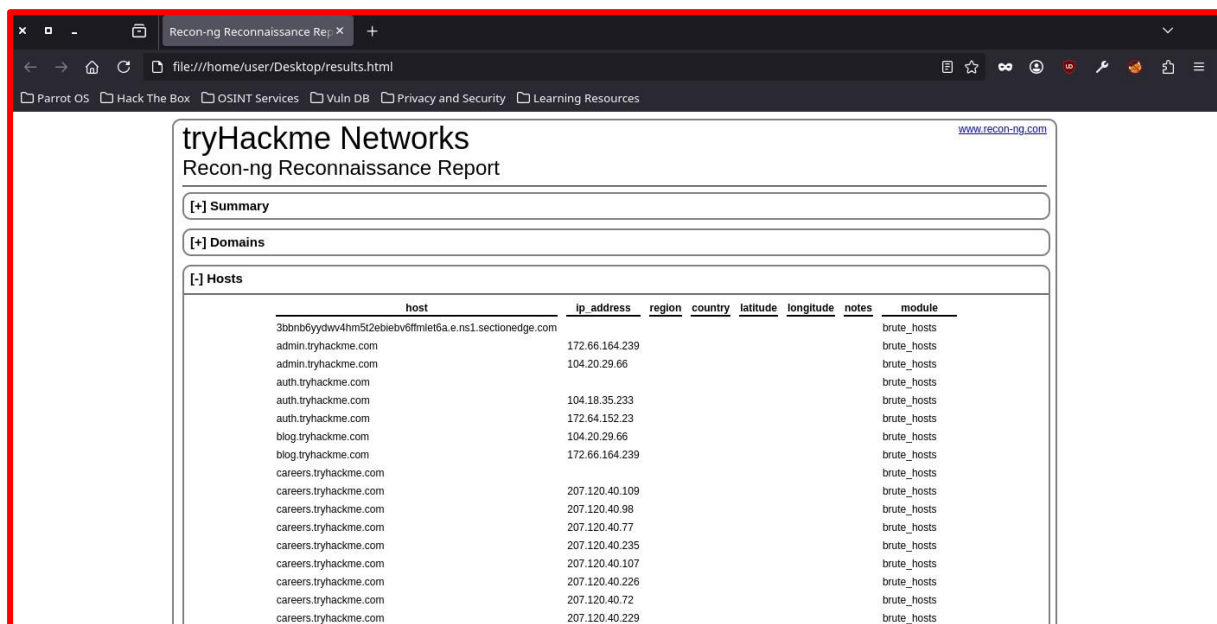
[*] No Wildcard DNS entry found.
[*] 02.tryhackme.com => No record found.
[*] 01.tryhackme.com => No record found.
[*] 0.tryhackme.com => No record found.
[*] 03.tryhackme.com => No record found.
```

Step 11: View Generated Report

Navigate to the desktop directory and open the generated **results.html** file using Firefox.



The report contains a structured summary of the discovered hosts and reconnaissance data.



11

Conclusion

Conclusion

This laboratory exercise provided a comprehensive understanding of **footprinting as the initial stage of ethical hacking and penetration testing**. Through various passive and semi-passive techniques, significant information about a target organization can be collected without directly interacting with internal systems.

Throughout the lab, several tools and techniques were explored, including:

- *Search engine reconnaissance*
- *Social networking analysis*
- *Whois and DNS lookup*
- *Network tracing*
- *Email header investigation*
- *Automated reconnaissance tools*

These activities demonstrated how publicly accessible data can reveal important information about an organization's infrastructure, employees, and network architecture.

Understanding footprinting techniques enables cybersecurity professionals to think from an attacker's perspective. By identifying exposed information and potential vulnerabilities, organizations can strengthen their security posture and protect themselves against real-world cyber threats.

References

EC-Council (2023). **Certified Ethical Hacker (CEH) Official Curriculum**

<https://www.eccouncil.org/>

Weidman, G. (2014). **Penetration Testing: A Hands-On Introduction to Hacking**

<https://nostarch.com/pentesting>

Kennedy, D., O’Gorman, J., Kearns, D., & Aharoni, M. (2018).

Metasploit: The Penetration Tester’s Guide

<https://nostarch.com/metasploit>

McClure, S., Scambray, J., & Kurtz, G. (2018).

Hacking Exposed: Network Security Secrets & Solutions

Stuttard, D., & Pinto, M. (2011).

The Web Application Hacker's Handbook

Bazzell, M. (2023).

Open Source Intelligence Techniques

OWASP Testing Guide

<https://owasp.org/>

SANS Institute Resources

<https://www.sans.org/>

